

SUPPORT, SCREENING, AND NEGATIVE GROWTH OF REFLECTED EXPONENTIAL FORMS

R. ARUN CHANDRU

ABSTRACT. We study compactly supported negative tests for exponential forms associated with reflected divisors in a vertical strip. For a fixed polynomial counting budget, we determine the order of the worst support required to attain a prescribed normalized negative margin in the presence of an off-axis pair. The estimate is uniform over arbitrary additional reflected pairs and infinite backgrounds, without separation between distinct locations. A centered conjugation symmetry changes the exponent through the parity of the local counting budget. We also derive an explicit small-displacement transition for finite screens and prove its stability under a fixed polynomial-count axis background. For a fixed divisor, negative values have a lower exponential growth rate determined by the outer displacement; a weighted local condition gives the matching upper rate. Finally, we construct locally semibounded forms with fixed sparse and logarithmic counting bounds whose negative envelopes dominate any prescribed growth. These results concerning abstract reflected forms separate support, normalized margin, and large-window growth.

CONTENTS

1. Support, margin, and growth	2
1.1. The uniform support law	2
1.2. A finite transition and two infinite-divisor conclusions	3
1.3. Relation to prior work and the contribution here	4
2. Reflected forms and compact tests	5
2.1. Changing the center and the scale	6
2.2. Finite features and their inertia	6
3. The sharp support–margin law at fixed count budget	6
3.1. Three ordinate bands	7
3.2. Grouping close nodes without a separation assumption	7
3.3. Selecting the tail budget before interpolation	9
3.4. Signs, norm, and physical support	9
3.5. The power branch and the logarithmic branch	10
3.6. A matching obstruction for every test	11
3.7. The exact role of centered conjugation symmetry	12
3.8. Fixed continuous counting profiles	12
4. Exact finite screening and fixed-background transitions	12
4.1. The coefficient and the two asymptotic conclusions	13
4.2. Scaling, parity, and exact inertia	14
4.3. The signed moment matrix and its Schur complement	14
4.4. A nonsingular feature basis and the eigenvalue expansion	16

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4.5.	Smooth finite-dimensional recovery of the coefficient	18
4.6.	The fixed axis background vanishes on the scaled tests	19
4.7.	Inversion to the first prescribed-margin radius	20
5.	Amplification and growth for one fixed divisor	20
5.1.	Fixed-target amplification with arbitrarily small relative support loss	21
5.2.	A weighted local condition sufficient for a finite envelope	22
5.3.	Exact exponential rate and the first prescribed-margin radius	23
6.	Arbitrary growth under fixed counting bounds	24
6.1.	Separating two radii with a finite screen	24
6.2.	A symmetric induction controlling both directions of interference	25
6.3.	Counts, including partially entered bands	26
6.4.	Absolute convergence and local semiboundedness	27
6.5.	Survival of the prescribed witnesses	27
7.	The arithmetic interface	28
	Acknowledgments	29
	Note on AI Use	30
	References	30

1. SUPPORT, MARGIN, AND GROWTH

A negative direction can exist and yet be difficult to resolve. For a reflected exponential form, a pair away from the imaginary axis creates an indefinite contribution. Other locations can mask that contribution on a short interval, or leave a negative value whose size is much smaller than the norm of its test. This leads to three different questions: when does negativity first occur; how much support is needed for a specified normalized margin; and how rapidly can the negative envelope grow for one fixed infinite configuration?

The distinction is already visible in a single pair. If $H_f(z) = \int f(u)e^{zu} du$, the pair at $\beta, -\beta$ contributes $2\Re(H_f(\beta)\overline{H_f(-\beta)})$. On $(-L, L)$, its largest normalized negative value is

$$\beta^{-1}\{\sinh(2\beta L) - 2\beta L\}.$$

It is positive for every $L > 0$, but for $\beta L \ll 1$ it is approximately $(4/3)\beta^2 L^3$. Thus existence of a negative test and existence of a unit-margin test are already different assertions before a background is added.

We study these questions for locally finite divisors Z in $|\Re z| \leq a$, invariant under $z^\# = -\bar{z}$, with positive integer multiplicities and a polynomial bound on the number of locations counted with multiplicity. The form is

$$Q_Z(f, g) = \sum_{z \in Z} \nu(z) H_f(z) \overline{H_g(z^\#)}.$$

We work throughout with actual compact smooth tests. Write $\mathcal{M}_Z(L)$ for the supremum of the positive part of $-Q_Z(f, f)/\|f\|_2^2$ on $(-L, L)$, and $r_Z(\eta)$ for the infimum of radii admitting a test with margin at least η . The definitions, including infinite envelopes and nonattained infima, are given in Section 2.

1.1. The uniform support law. Our first main result fixes the counting parameters and varies the displacement and requested margin. Suppose

$$N_0(Y) = \sum_{|\Im z| \leq Y} \nu(z) \leq C(1 + Y)^m \quad (Y \geq 0), \quad C \geq 2,$$

and require $\beta, -\beta \in Z$. All other reflected pairs are permitted. Let $\mathcal{R}_\eta(C, \beta; a, m)$ be the worst value of $r_Z(\eta)$ over this entire class. For fixed a, C, m , we prove

$$(1.1) \quad \mathcal{R}_\eta(C, \beta; a, m) \asymp_{a, C, m} \beta^{-1} \log \left(1 + (\beta\eta)^{1/(2\lfloor C \rfloor - 1)} \right), \quad \eta \geq 1, \quad 0 < \beta \leq \min(a, 1).$$

This is an order estimate, not an exact leading minimax constant. At unit margin the support exponent is $(2\lfloor C \rfloor - 2)/(2\lfloor C \rfloor - 1)$. The witness may depend on the entire divisor. The conclusion does not provide a predetermined finite collection of tests.

The local integer budget determines the exponent, while the rest of the counting bound enters the constants. Arbitrarily close distinct nodes prevent a naive interpolation estimate based on inverse separation. We instead prescribe opposite values on two reflected groups of nearby nodes, annihilate an intermediate band, and suppress the full remaining tail. The grouped contour length saves one power in the coefficient estimate. Applying the intermediate polynomial as a differential operator in physical coordinates, before a probability convolution, preserves the needed norm bound. Finite unit-weight screens give the matching obstruction against every test.

An additional conjugation symmetry about the target ordinate changes the exponent denominator to $4\lfloor C/2 \rfloor - 1$. This is a parity effect of the integer count: an origin atom is invisible to an odd test and the remaining local mass is even. It is not legitimate to infer the centered symmetry from global conjugation after translating a nonzero target height. Section 3 proves both versions with their full parameter ranges.

1.2. A finite transition and two infinite-divisor conclusions. Finite screens make the mechanism more explicit. For the Chebyshev screen

$$\mathcal{S}_{n, \beta} = 128 \sum_{j=1}^n \left[i\beta n \cos \frac{(2j-1)\pi}{2n} \right] + [\beta] + [-\beta], \quad d = 2\lfloor n/2 \rfloor + 1,$$

we compute the unique negative eigenvalue and obtain

$$\mathcal{M}_{\mathcal{S}_{n, \beta}}(L) = c_n \beta^{2d} L^{2d+1} (1 + O_n((\beta L)^2)), \quad c_n > 0.$$

Here n is fixed and $\beta L \rightarrow 0$. For any fixed polynomial-count axis divisor A , the leading transition survives:

$$(1.2) \quad \mathcal{M}_{\mathcal{S}_{n, \beta} + A}(x\beta^{-q}) \longrightarrow c_n x^{2d+1}, \quad q = \frac{2d}{2d+1}, \quad \beta \downarrow 0,$$

locally uniformly for $x > 0$. The background may define an unbounded positive form. It is held fixed, and the window expands; neither qualification can be omitted. The proof in Section 4 uses finite smooth moment recovery followed by a separate high-frequency estimate on that fixed space.

For a fixed divisor with a nonempty off-axis part, put $s_Z = \sup\{\Re z : z \in Z, \Re z > 0\}$. Polynomial counting alone gives

$$(1.3) \quad \liminf_{L \rightarrow \infty} \frac{\log \mathcal{M}_Z(L)}{L} \geq 2s_Z.$$

The logarithm is interpreted in the extended sense if an envelope is infinite. If, in addition,

$$B_Z = \sup_{y \in \mathbb{R}} \sum_{\substack{z = \sigma + i\gamma \in Z \\ \sigma > 0, |\gamma - y| \leq 1}} \nu(z) \sigma^2 < \infty,$$

then $\log \mathcal{M}_Z(L)/L \rightarrow 2s_Z$, and $r_Z(\eta) \sim (2s_Z)^{-1} \log \eta$ as $\eta \rightarrow \infty$. Section 5 proves the lower construction in every sufficiently large window and the upper estimate by a weighted area-mean argument. The condition on B_Z is sufficient, not a characterization of local semiboundedness.

Our second main conclusion shows how far local semiboundedness alone is from a growth law. Given any finite-valued positive nondecreasing G on $[2, \infty)$, we construct one divisor, invariant under both reflection and conjugation, in $|\Re z| \leq 1$, for which

$$(1.4) \quad \begin{aligned} N_0(Y) &\leq Y + 2, & N_t(1) &\leq 2 + \log(2 + |t|), \\ \mathcal{M}_Z(L) &< \infty \quad (0 < L < \infty), & \mathcal{M}_Z(L) &\geq G(L) \quad (L \geq 2). \end{aligned}$$

The constants in the count bounds do not depend on G . The construction separates two radii with a finite screen and then places conjugate pairs of screens far apart in height. It controls both the effect of each new component on old witnesses and the effect of old components on the new witness. Summable small-window negative bounds coexist with the prescribed large-window witnesses.

There is no conflict between (1.1), (1.2), and (1.4). The first takes a supremum over varying divisors with fixed counting parameters; the second holds one axis background fixed while a displacement tends to zero; the third fixes a specially constructed infinite divisor and lets support grow. Keeping these quantifiers visible is central to the subject.

1.3. Relation to prior work and the contribution here. The arithmetic motivation is Weil’s positivity criterion, but the forms in this paper are abstract. Bombieri [3, Sections 4, 8–11] studies compact-support variational problems for the actual Weil functional and finite exponential-kernel matrices. In particular, his finite-matrix analysis determines negative inertia from off-axis pairs and discusses the delicate passage to infinite configurations. We use the elementary finite inertia mechanism as background, not as a new result. Our uniform result specifies the joint support and L^2 -margin cost over a polynomial-count class, allowing an arbitrary additional off-axis part and controlling its entire tail. It is not an arithmetic strengthening of Bombieri’s conclusions.

The collision powers in finite exponential systems also have substantial precedents in super-resolution. Demanet and Nguyen [5, Theorem 2 and Corollary 3] obtain sharp small-bandwidth powers for sparse recovery on a grid. Batenkov, Demanet, Goldman, and Yomdin [1, Theorems 3.2 and 3.9] give lower bounds for nonuniform Fourier matrices under specified clustering conditions, together with examples having the matching cluster-size power. Those are singular-value and recovery problems. Here the quantity is an indefinite, normalized quadratic value, and the upper construction must preserve signs in an infinite complex reflected configuration. The appearance of an odd power is not itself a novelty.

Likewise, the moment and Schur-complement calculations used for a finite screen belong to established perturbation theory. Barthelmé and Usevich [2] treat kernel matrices in the flat limit, and Usevich and Barthelmé [7] treat analytic symmetric perturbations, including indefinite matrices. The contribution of our finite-screen analysis is the explicit negative-margin transition and the fixed infinite-background limit (1.2), not a new general matrix-perturbation method. We include the calculation needed here rather than require a general perturbation theorem. Classical interpolation and orthogonal-polynomial normalizations are recorded in [6].

Finite zero-annihilating products multiplied by high powers of sinc also occur in Zhu’s RH-conditional construction [8, Section 13.2]. That product-window template is not claimed as novel in this work. The sign-compatible interpolation, physical norm estimate, and full-tail bounds required for our indefinite problem are proved below; none of that preprint’s other arithmetic assertions is used.

The principal conclusions established here are therefore the full-class fixed-count law (1.1), its centered-symmetry refinement, the stated fixed-background transition, and the separation between local semiboundedness and arbitrary growth (1.4). The fixed-divisor growth estimates connect these conclusions and give a useful sufficient class in which the outer displacement is recoverable from the exponential rate.

Recent work on compact Weil operators and near-radical vectors, for example Connes and Consani [4], concerns additional arithmetic structure that our abstract divisor class does not possess. The final section gives the actual prime, gamma, and polar expression in our normalization and explains what an arithmetic upper bound would imply. We prove no such new upper bound here.

2. REFLECTED FORMS AND COMPACT TESTS

Throughout, a divisor is a locally finite multiset of complex numbers. We sum over its distinct locations and denote their positive integer multiplicities by $\nu(z)$. Fix a strip half-width $a > 0$, and assume

$$(2.1) \quad |\Re z| \leq a, \quad z^\# = -\bar{z}, \quad z \in Z \iff z^\# \in Z, \quad \nu(z) = \nu(z^\#).$$

Unless explicitly restricted to a finite divisor, we assume a polynomial ordinate count: there are finite $C > 0, m \geq 0$ such that

$$(2.2) \quad N_0(Y) := \sum_{|\Im z| \leq Y} \nu(z) \leq C(1+Y)^m \quad (Y \geq 0).$$

The constants need not be uniform over all divisors except in the uniform assertions where this is stated.

For $f, g \in C_c^\infty(\mathbb{R}; \mathbb{C})$, set

$$(2.3) \quad H_f(z) = \int_{\mathbb{R}} f(u) e^{zu} du, \quad Q_Z(f, g) = \sum_{z \in Z} \nu(z) H_f(z) \overline{H_g(z^\#)}.$$

Our inner products are linear in the first variable. Repeated integration by parts gives, for every integer $j \geq 0$,

$$\sup_{|\sigma| \leq a} |H_f(\sigma + it)| \leq C_{f,j,a} (1 + |t|)^{-j}.$$

For $|t| \geq 1$, this follows by integrating the Fourier transform of $e^{\sigma u} f(u)$ by parts j times, uniformly in the bounded real parameter; bounded t is immediate. Dyadic summation using (2.2) proves absolute convergence of (2.3). Changing z to $z^\#$ then proves $Q_Z(f, g) = \overline{Q_Z(g, f)}$.

For $z = \sigma + i\gamma$ with $\sigma > 0$, write

$$A_z(f) = \frac{H_f(\sigma + i\gamma) + H_f(-\sigma + i\gamma)}{2}, \quad D_z(f) = \frac{H_f(\sigma + i\gamma) - H_f(-\sigma + i\gamma)}{2}.$$

The two locations of this pair contribute exactly

$$(2.4) \quad 2\nu(z) (|A_z(f)|^2 - |D_z(f)|^2).$$

An axis location $i\gamma$ contributes $\nu(i\gamma) |H_f(i\gamma)|^2 \geq 0$. Formula (2.4) explains both the possibility of a negative test and the possibility that positive terms screen it. Multiplicity weights a value; it does not ask the transform to vanish to higher order.

The quantities studied below are

$$(2.5) \quad \mathcal{M}_Z(L) = \sup_{0 \neq f \in C_c^\infty(-L, L)} \left(-Q_Z(f, f) / \|f\|_2^2 \right)_+, \quad L > 0,$$

$$r_Z(\eta) = \inf \{ L > 0 : \exists 0 \neq f \in C_c^\infty(-L, L), Q_Z(f, f) \leq -\eta \|f\|_2^2 \}, \quad \eta > 0.$$

Here $x_+ = \max(x, 0)$, the first supremum may be infinite, and the infimum of the empty set is infinite. No extremum is assumed attained. In particular, $\mathcal{M}_Z(L) > \eta$ supplies an actual test with margin exceeding η , whereas equality alone need not. Enlarging L enlarges the test class, so $\mathcal{M}_Z$ is nondecreasing. Adding an axis divisor can only decrease this envelope. We say that the

form is locally semibounded when $\mathcal{M}_Z(L) < \infty$ for every finite $L > 0$; this terminology asserts a bound on the smooth test class, not a closed operator realization.

2.1. Changing the center and the scale. For a target ordinate γ , use the centered count

$$N_\gamma(Y) = \sum_{|\Im z - \gamma| \leq Y} \nu(z).$$

If $W = Z - i\gamma$ and $f(u) = e^{-i\gamma u}v(u)$, then $H_f(z) = H_v(z - i\gamma)$ and $Q_Z(f, f) = Q_W(v, v)$. Norm and support are unchanged. Thus a target can be moved to the real axis without a height-dependent cost in the test norm. An origin-centered count (2.2) implies the centered bound

$$N_\gamma(Y) \leq C(1 + |\gamma|)^m(1 + Y)^m.$$

The constant here depends on the chosen target.

For $\alpha > 0$, let $W = Z/\alpha$, with the same multiplicities, and put $f(u) = \sqrt{\alpha}v(\alpha u)$. Direct substitution yields

$$(2.6) \quad \|f\|_2 = \|v\|_2, \quad H_f(z) = \alpha^{-1/2}H_v(z/\alpha), \quad Q_Z(f, f) = \alpha^{-1}Q_W(v, v).$$

A radius L for v becomes L/α for f . These factors will matter when passing from a normalized strip to a growth rate.

2.2. Finite features and their inertia. For a finite divisor, (2.3) extends continuously to $L^2(-L, L)$. List its distinct locations as $z_1, \dots, z_N$, and define $(Ef)_j = \sqrt{\nu(z_j)}H_f(z_j)$. Let J permute coordinates according to reflection. Then

$$(2.7) \quad Q_Z(f, f) = \langle Ef, JEf \rangle_{\mathbb{C}^N}, \quad A_Z = E^*JE.$$

The exponential functions representing the evaluations are linearly independent on every nonempty interval. Indeed, a vanishing finite linear combination is analytic and hence vanishes identically; its first N derivatives at any point give an invertible Vandermonde system. Thus E is onto. On the orthogonal complement of its kernel, (2.7) is a congruence of J . There is one negative direction for each distinct off-axis reflected pair, one positive direction for each such pair, and one positive direction for each distinct axis location. All remaining directions are null. This finite-inertia mechanism is classical; compare Bombieri's finite exponential-kernel analysis [3, Sections 8–9].

Smooth compact functions are dense in $L^2(-L, L)$, and the finite form is continuous. Consequently its least Rayleigh value and negative envelope can be computed on either class. A strict inequality found in L^2 survives smooth approximation. For an infinite axis background the form need not be bounded, so this density argument cannot be used without additional estimates. The fixed-background argument below supplies those estimates.

Finally, for the single unit pair $[\beta] + [-\beta]$, (2.4) and the even/odd decomposition give

$$(2.8) \quad \mathcal{M}_{[\beta] + [-\beta]}(L) = 2 \|\sinh(\beta u)\|_{L^2(-L, L)}^2 = \beta^{-1} \{\sinh(2\beta L) - 2\beta L\}.$$

This exact elementary example has size $(4/3)\beta^2 L^3$ when βL tends to zero and size $e^{2\beta L}/(2\beta)$ when βL tends to infinity. The question is how much these scales can be delayed by a counting budget and a positive background.

3. THE SHARP SUPPORT-MARGIN LAW AT FIXED COUNT BUDGET

Fix $a > 0$, $C \geq 2$, and $m \geq 0$. For $0 < \beta \leq \min(a, 1)$, let $\mathcal{Z}(C, \beta; a, m)$ be the class of divisors satisfying (2.1), containing β and $-\beta$, and obeying $N_0(Y) \leq C(1 + Y)^m$ for every $Y \geq 0$. Define

$$\mathcal{R}_\eta(C, \beta; a, m) = \sup_{Z \in \mathcal{Z}(C, \beta; a, m)} r_Z(\eta).$$

Other off-axis pairs are unrestricted except for the strip, reflection, and count. The divisor, not only the test, varies in this supremum.

Put $M = \lfloor C \rfloor$, $p = 2M - 1$. The uniform conclusion is

$$(3.1) \quad \mathcal{R}_\eta(C, \beta; a, m) \asymp_{a, C, m} \beta^{-1} \log(1 + (\beta\eta)^{1/p}), \quad \eta \geq 1, \quad 0 < \beta \leq \min(a, 1).$$

Both comparison constants are positive and finite. They are uniform in β, η, Z , but no uniformity as C grows is asserted. In particular,

$$\mathcal{R}_1(C, \beta; a, m) \asymp_{a, C, m} \beta^{-(2M-2)/(2M-1)}.$$

We will prove the upper bound by a smooth interpolation construction and the lower bound against every test by a finite screen. A convenient equivalent comparison function is $\beta^{-1}\Phi_p(\beta\eta)$, where

$$(3.2) \quad \Phi_p(t) = \begin{cases} t^{1/p}, & 0 < t \leq 1, \\ \log(1+t), & t > 1. \end{cases}$$

For fixed p , this is comparable with $\log(1+t^{1/p})$ on the whole positive half-line.

3.1. Three ordinate bands. Choose $0 < \delta \leq 1$ so that $C(1+\delta)^m < M+1$. Such a choice exists by continuity; if $m = 0$, take $\delta = 1$. Integer multiplicities imply $N_0(\delta) \leq M$. List the distinct low locations as $z_1, \dots, z_k$, so $2 \leq k \leq M$.

Choose any fixed $R \geq 16(a+2)$, and put

$$d_0 = 4/R, \quad r_* = \lceil C(1+R)^m \rceil, \quad V = a + R.$$

Let $v_1, \dots, v_r$ be the distinct intermediate locations, with $\delta < |\Im v_j| \leq R$; then $r \leq r_*$. The intermediate annihilator and two useful bounds are

$$(3.3) \quad A(w) = \prod_{j=1}^r (1 - w/v_j), \quad A_0 = (1 + \delta^{-1})^{r_*}, \quad A_1 = (2V/\delta)^{r_*}.$$

The coefficient sum of A is at most A_0 . If $|\Im w| \leq \delta/2$, then $|1 - w/v_j| > \delta/(2V)$, so $|A(w)|^{-1} \leq A_1$. The empty product is one. Every actual intermediate location is annihilated, irrespective of its multiplicity.

Fix an even nonnegative $h \in C_c^\infty(-1, 1)$ with $\int h = 1$, and write $G = H_h$. For $|\Im s| \leq 1/4$,

$$(3.4) \quad \Re G(s) \geq \cos(1/4)G(\Re s) \geq \cos(1/4) > 1/2.$$

The second inequality follows from evenness and $\cosh x \geq 1$. Thus the reciprocal of G is holomorphic and bounded by two in the narrower strip in which we will interpolate.

3.2. Grouping close nodes without a separation assumption. For now take free parameters

$$(3.5) \quad \ell \geq 1, \quad \epsilon = \beta\ell \leq 1, \quad \tau \geq 0, \quad T = \tau\ell.$$

Scale the low nodes to $s_j = \ell z_j$, including $\epsilon, -\epsilon$, and set

$$\rho_0 = \frac{\delta\epsilon}{64M(1+\tau)}.$$

Choose $\rho \in [\rho_0, 2\rho_0]$ avoiding the finitely many radii for which two boundary circles are tangent or three distinct boundary circles meet. Form the union of the disks $|s - s_j| < \rho$, and let Ω_+, Ω_- be the components containing $\epsilon, -\epsilon$.

A path in the disk-intersection graph uses at most $k-1$ edges without repetition. Consequently every point of either selected component's closure is within $2k\rho$ of its target, and

$$(3.6) \quad 2k\rho \leq \frac{\delta\epsilon}{16(1+\tau)}, \quad |\Im s| \leq \frac{\delta\epsilon}{16(1+\tau)}, \quad |s| < 2 \quad (s \in \overline{\Omega}_+ \cup \overline{\Omega}_-).$$

The components are distinct, lie in opposite open half-planes, and are interchanged by $s \mapsto -\bar{s}$. Orient their outer boundaries positively and any hole boundaries negatively. They are finite piecewise smooth contours with total length at most $2\pi k\rho$. Every boundary point has distance at least ρ from every low node. A node in a hole is therefore excluded with winding number zero, as it should be.

Normalize the nodal polynomial by

$$p_0(s) = \prod_{j=1}^k \frac{s - s_j}{\sqrt{1 + |s_j|^2}}.$$

At a selected boundary point ζ , the normalized distance to a node of modulus at most four is at least $\rho/\sqrt{17}$. For a larger node, $|\zeta - s_j| > |s_j|/2$, giving a lower bound $1/\sqrt{5}$. Thus

$$(3.7) \quad |p_0(\zeta)| \geq (\rho/5)^k, \quad \left\| \frac{p_0(\zeta) - p_0(s)}{\zeta - s} \right\|_{\text{coef},1} \leq k3^{k-1}.$$

Here $\|\cdot\|_{\text{coef},1}$ denotes the sum of absolute coefficients. To see the second estimate, telescope the difference of the two products. A polynomial factor has coefficient sum at most $\sqrt{2}$, an evaluated factor has modulus at most $\sqrt{5}$, and the removed linear factor contributes at most one. Bounding the remaining $k-1$ factors by three proves the assertion. This is the point at which normalizing far-away scaled nodes is useful.

Let $b(w) = \sinh(d_0 w)/(d_0 w)$, with its value one at zero. For real x, y , elementary inequalities give

$$\left| \frac{\sinh(x + iy)}{x + iy} \right|^2 \geq \left(\frac{\sin y}{y} \right)^2, \quad \left| \frac{\sinh(x + iy)}{x + iy} \right|^{-K} \leq e^{2Ky^2} \quad (|y| \leq 1/4).$$

Indeed $\sinh^2 x \geq x^2$, $\sin^2 y \leq y^2$, and $\sin y/y \geq 1 - y^2/6$; taking logarithms gives the second bound. By (3.6), on the selected closures

$$|b(s/\ell)|^{-K} \leq e^{\chi K}, \quad \chi = \frac{\delta^2}{8R^2} \frac{\beta^2}{(1 + \tau)^2}.$$

For any positive integer K , the data

$$M_K(s) = \{A(s/\ell)b(s/\ell)^K G(s)\}^{-1}$$

are holomorphic near both selected closures and bounded in modulus by $2A_1 e^{\chi K}$. Nonvanishing holds on containing small target disks, so possible holes cause no analytic difficulty.

The polynomial

$$(3.8) \quad P(s) = \frac{1}{2\pi i} \left(\int_{\partial\Omega_+} - \int_{\partial\Omega_-} \right) M_K(\zeta) \frac{p_0(\zeta) - p_0(s)}{p_0(\zeta)(\zeta - s)} d\zeta$$

has degree at most $k-1$. Cauchy's formula gives the values $+M_K(s_j)$, $-M_K(s_j)$, or zero, according as the node is in Ω_+ , Ω_- , or neither. The length bound and (3.7) give

$$(3.9) \quad \|P\|_{\text{coef},1} \leq D_0 \epsilon^{1-k} (1 + \tau)^{k-1} e^{\chi K}, \quad D_0 = 2A_1 M^2 3^{M-1} 5^M (64M/\delta)^{M-1}.$$

The boundary length supplies one power of ρ . Losing that power would give the wrong exponent in (3.1). No inverse minimum separation has appeared.

3.3. Selecting the tail budget before interpolation. We now choose K , rather than assume that a later choice can absorb the size of the interpolant. All of the following are fixed once a, C, m, R are fixed:

$$\begin{aligned} D_{\text{tail}} &= D_0(2R)^{M-1}(3R/\delta)^{r_*}, \\ S_R &= 2^{m+1}C(1+R)^m, \quad B_R = \tfrac{1}{2}\log(2S_R), \\ d_* &= M + r_* + m, \quad J_* = 1 + d_* + \log D_{\text{tail}} + B_R. \end{aligned}$$

Put $u_\beta = \log(1/\beta)$, and select

$$\begin{aligned} E &= a\ell(1+\tau) + (M-1)\{u_\beta + \log(1+\tau)\} + \log D_{\text{tail}}, \\ (3.10) \quad K &= \lceil \max\{d_*, E + B_R\} \rceil \leq a\ell(1+\tau) + (M-1)\{u_\beta + \log(1+\tau)\} + J_*. \end{aligned}$$

Use this integer in (3.8). The complete transform will be

$$(3.11) \quad \mathcal{F}(w) = \sqrt{\ell} P(\ell w) G(\ell w) A(w) b(w)^K \cosh(Tw).$$

Before the final cosh factor, its low-node values are $+\sqrt{\ell}, -\sqrt{\ell}, 0$, as prescribed, and all its intermediate values vanish.

For $w = \sigma + it$, $|\sigma| \leq a$, $v = |t| > R$, we have

$$\begin{aligned} 1 + \ell|w| &\leq 2\ell v, & |A(w)| &\leq (3R/\delta)^{r_*}(v/R)^r, \\ |G(\ell w)| &\leq e^{a\ell}, & |\cosh(Tw)| &\leq e^{a\tau\ell}, \\ |b(w)| &\leq e^{4a/R} \frac{R}{4v}. \end{aligned}$$

In the polynomial bound the factors of ℓ cancel exactly: $\epsilon^{1-k}(2\ell v)^{k-1} = \beta^{1-k}(2v)^{k-1}$. Moreover $\chi \leq 1/16$, $4a/R < 1/4$, and hence $\log 4 - 4a/R - \chi > 1$. Substituting (3.9) into (3.11) therefore gives

$$(3.12) \quad |\mathcal{F}(\sigma + it)| \leq \sqrt{\ell} e^{E-K} (v/R)^{-q_*}, \quad q_* = K - (k-1) - r \geq m+1.$$

Dyadic summation gives $\sum_{|\Im z| > R} \nu(z) (|\Im z|/R)^{-2q_*} \leq S_R$. Apply (3.12) at both reflected locations to obtain

$$(3.13) \quad \sum_{|\Im z| > R} \nu(z) |\mathcal{F}(z) \overline{\mathcal{F}(z^\#)}| \leq \ell e^{2(E-K)} S_R \leq \ell/2.$$

The estimate concerns the full reflected tail, not a diagonal substitute for it.

3.4. Signs, norm, and physical support. Reflection exchanges the selected components. A selected pair at z contributes $-2\nu(z)\ell \Re \cosh^2(Tz)$. By (3.6), $|T\Im z| \leq 1/16$. The identity

$$\Re \cosh^2(x + iy) = \tfrac{1}{2}\{1 + \cosh(2x) \cos(2y)\}$$

shows that every selected pair contributes negatively. Keeping the exact contribution of the targets until after tail absorption gives

$$(3.14) \quad Q_Z(f, f) \leq -2\ell \cosh^2(\tau\epsilon) + \ell/2 \leq -\tfrac{1}{2}\ell e^{2\tau\epsilon}.$$

All other low-node contributions and all intermediate contributions are zero.

Here the inverse transform is an actual smooth test. Let μ be uniform probability measure on $[-d_0, d_0]$, and put

$$h_\ell(u) = \ell^{-1/2} h(u/\ell), \quad f_0 = \mu^{*K} * \{A(-\partial_u)P(-\ell\partial_u)h_\ell\}, \quad f(u) = \tfrac{1}{2}\{f_0(u-T) + f_0(u+T)\}.$$

Integration by parts and Fubini give $H_f = \mathcal{F}$. Differentiation preserves the initial bump's strict support gap, and convolution by a compact measure preserves smoothness. Thus

$$(3.15) \quad f \in C_c^\infty(-L_{\text{out}}, L_{\text{out}}), \quad L_{\text{out}} = \ell(1+\tau) + 4K/R.$$

The order of operations yields an efficient norm estimate. A term of degree j from P and degree b from A acts on the bump as a scalar multiple of $\ell^{-b}\ell^{-1/2}h^{(j+b)}(u/\ell)$. Its norm is at most $\|h^{(j+b)}\|_2$, since $\ell \geq 1$. Probability convolution and the average of two translations are L^2 contractions. Hence, with $H_* = \max_{0 \leq j \leq M-1+r_*} \|h^{(j)}\|_2$,

$$(3.16) \quad \|f\|_2 \leq A_0 H_* \|P\|_{\text{coef},1}.$$

In particular, the convolution incurs no exponential norm cost in K , and the intermediate differential operator incurs no positive power of ℓ .

Write $\mu_0 = \delta^2/(8R^2)$. Multiplying (3.10) by $\chi = \mu_0\beta^2/(1+\tau)^2$, and using $\beta\ell \leq 1$, gives

$$\chi K \leq \mu_0 \{a + (M-1)/e + J_*\} =: H_{\text{inv}}.$$

Here we used $\beta^2 \log(1/\beta) \leq 1/(2e)$ and $\log(1+\tau)/(1+\tau)^2 \leq 1/(2e)$. Let

$$B = \max\{1, (A_0 H_* D_0)^2 e^{2H_{\text{inv}}}\}.$$

Equations (3.9) and (3.16) then give

$$\|f\|_2^2 \leq B \left(\frac{1+\tau}{\epsilon} \right)^{2M-2}.$$

The negative value proves $f \neq 0$. We have proved the following construction for every set of parameters in (3.5):

$$(3.17) \quad -\frac{Q_Z(f, f)}{\|f\|_2^2} \geq c\ell \left(\frac{\beta\ell}{1+\tau} \right)^{p-1} e^{2\tau\beta\ell},$$

$$L_{\text{out}} \leq D\{\ell(1+\tau) + \log(1/\beta) + 1\},$$

where $c = 1/(2B) > 0$ and a finite D depend only on a, C, m, R, h . For the second line, use $\log(1+\tau) \leq \tau$, $\ell \geq 1$ in (3.10). For example $D = 1 + 4a/R + 4(M-1)/R + 4J_*/R$ suffices. We have retained (3.10) and (3.15) separately because their leading dependence on T will yield an exact growth exponent later.

3.5. The power branch and the logarithmic branch. For (3.1) fix $R = 16(a+2)$. Let $t = \beta\eta$, $q = (p-1)/p$, choose $A \geq 1$ with $cA^p \geq 2$, and put $t_0 = A^{-p}$. If $0 < t \leq t_0$, set

$$\ell = A\beta^{-1}t^{1/p} = A\eta^{1/p}\beta^{-q}, \quad \tau = 0.$$

Then $\ell \geq 1$, $\beta\ell \leq 1$, and the margin in (3.17) is at least 2η . The elementary maximum of $x^q \log(1/x)$ gives $\log(1/\beta) \leq \beta^{-q}/(qe) \leq \ell/(qe)$. Since $q \geq 2/3$, the support is at most $3DA\beta^{-1}t^{1/p}$.

For $t > t_0$, take $\ell = 1/\beta$, so $\epsilon = 1$. Choose

$$H = \max\{1, 4(p-1)^2, 2\log(2/c)\}, \quad K_1 = \max\{1, H/\log 2\}, \quad \tau = K_1 \log(2+t).$$

For $\tau \geq H$, the inequality $\log(1+\tau) \leq \sqrt{\tau}$ implies $(p-1)\log(1+\tau) \leq \tau/2$. One direct verification of the logarithmic inequality differentiates $\sqrt{x} - \log(1+x)$, whose derivative is $(\sqrt{x}-1)^2/(2\sqrt{x}(1+x)) \geq 0$. It follows that $ce^\tau/(1+\tau)^{p-1} \geq 2$. The margin is therefore at least $2\beta^{-1}e^\tau > 2\eta$, while the support is bounded by a fixed constant times $\beta^{-1}\log(2+t)$. For $t \geq 1$, this is at most twice the logarithmic scale in (3.2). For the remaining interval $t_0 < t \leq 1$, use $\log(2+t) \leq \log 3 \leq A \log 3 t^{1/p}$. This covers the entire transition interval and proves the upper bound, always with a strict margin reserve.

3.6. A matching obstruction for every test. Let $n = M - 2$, so $p = 2n + 3$, and define

$$(3.18) \quad T_n^2 = (n+1)^3 4^n, \quad R_n = 24T_n^2, \quad Z_{n,\beta} = [\beta] + [-\beta] + \sum_{j=1}^n [i\beta R_n j].$$

The notation $[z]$ denotes one atom. This divisor has total mass $M \leq C$, and so satisfies the whole polynomial counting bound.

For every polynomial P of degree at most n , with coefficient vector b , we first claim

$$(3.19) \quad 2\Re(P(1)\overline{P(-1)}) + \sum_{j=1}^n |P(iR_n j)|^2 \geq (2T_n^2)^{-1} \|b\|_2^2.$$

Lagrange interpolation at $0, i, \dots, in$ gives basis polynomials of coefficient sum at most $(n+1)2^n$: the denominator at ij has modulus $j!(n-j)!$, while the numerator's coefficient sum is at most $(n+1)!$. Cauchy–Schwarz consequently gives

$$2|p(0)|^2 + \sum_{j=1}^n |p(ij)|^2 \geq T_n^{-2} \|\text{coef}(p)\|_2^2.$$

For $0 \leq h \leq 1/2$, a geometric-series estimate gives $|p(\pm h) - p(0)| \leq 2h \|\text{coef}(p)\|_2$. Replacing $2|p(0)|^2$ by $2\Re(p(h)\overline{p(-h)})$ changes the form by at most $12h \|\text{coef}(p)\|_2^2$. Take $h = 1/R_n$ and apply this to $p(w) = P(R_n w)$; its coefficient norm is at least that of P . This proves (3.19), including $n = 0$.

For $g \in L^2(-1, 1)$, Taylor expand

$$H_g(\epsilon s) = P_g(s) + r_g(s), \quad P_g(s) = \sum_{k=0}^n \frac{\epsilon^k s^k}{k!} \int_{-1}^1 x^k g(x) dx.$$

Set $K_0 = \max(1, nR_n)$. At each of the fixed nodes $1, -1, iR_n, \dots, inR_n$, the exponential remainder gives

$$|r_g(s)| \leq E_n \epsilon^{n+1} \|g\|_2, \quad E_n = \frac{\sqrt{2} e K_0^{n+1}}{(n+1)!}, \quad K_0 \epsilon \leq 1.$$

Let V_n evaluate coefficient vectors at these nodes and let J_0 swap the first two coordinates and fix the others. Its norm is one. With $b = \text{coef}(P_g)$, the polynomial part is at least $b_n \|b\|^2$, where $b_n = (2T_n^2)^{-1}$. The cross term is bounded by $2\|V_n\| \|b\| \|r\|$. Completing the square gives

$$(V_n b + r)^* J_0 (V_n b + r) \geq -(1 + \|V_n\|^2/b_n) \|r\|^2 \geq -D_n \epsilon^{2n+2} \|g\|_2^2,$$

with $D_n = (1 + \|V_n\|^2/b_n)(n+2)E_n^2$. Scaling from $(-1, 1)$ to $(-L, L)$ yields the all-test estimate

$$(3.20) \quad Q_{Z_{n,\beta}}(f, f) \geq -D_n L(\beta L)^{p-1} \|f\|_2^2 \quad (\beta L \leq K_0^{-1}).$$

This bounds the original form, not merely a polynomial compression.

For $t = \beta\eta \leq 1$, choose $k_1 > 0$ so that $k_1 \leq K_0^{-1}$, $D_n k_1^p \leq 1/2$, and take $L = k_1 \beta^{-1} t^{1/p}$. The envelope is at most $\eta/2$, excluding the requested margin at this radius and all smaller radii. For $t \geq 1$, axis positivity and (2.8) instead imply

$$\mathcal{M}_{Z_{n,\beta}}(L) \leq e^{2\beta L}/(2\beta).$$

At $L = (4\beta)^{-1} \log(1+t)$, its ratio to η is at most $\sqrt{1+t}/(2t) \leq \sqrt{2}/2 < 1$. Taking the smaller lower constant proves (3.1) in both regimes with the same finite screen.

3.7. The exact role of centered conjugation symmetry. If in addition $Z = \overline{Z}$ in the coordinates used above, let $\mathcal{R}_\eta^{\text{sym}}$ denote the supremum over this smaller class. Then

$$(3.21) \quad \mathcal{R}_\eta^{\text{sym}}(C, \beta; a, m) \asymp_{a, C, m} \beta^{-1} \log(1 + (\beta\eta)^{1/p_e}), \quad p_e = 4\lfloor C/2 \rfloor - 1.$$

Here too $\eta \geq 1$ and $0 < \beta \leq \min(a, 1)$.

To obtain the upper bound, omit zero from the low interpolation list. Every remaining location has its distinct negative with equal multiplicity, so the number of distinct nonzero low nodes is even and at most $M_e = 2\lfloor C/2 \rfloor$. Use M_e in all local degree and contour budgets, retaining the original count for the intermediate and tail bounds. The intermediate polynomial A is even and real, as are G and b . The selected positive component is conjugation-invariant and the negative component is its negative. The prescribed interpolation data are odd and conjugation-compatible. Uniqueness of polynomial interpolation therefore shows that P is odd and has real coefficients. The resulting physical test is real and odd. Its transform vanishes at zero, accounting for every omitted zero multiplicity. All estimates already proved hold with exponent $2M_e - 1 = p_e$.

For the lower bound write $M_e = 2r + 2$, and use

$$[\beta] + [-\beta] + \sum_{j=1}^r \{[i\beta R_r j] + [-i\beta R_r j]\}, \quad T_r^2 = (2r + 1)(r + 1)^{4r}, \quad R_r = 24T_r^2.$$

Lagrange interpolation on $-ir, \dots, 0, \dots, ir$ has coefficient bound T_r by the same argument: each numerator coefficient sum is at most $(r + 1)^{2r}$, and each denominator modulus is at least one. The perturbation from the repeated zero evaluations to $\pm 1/R_r$ costs at most $12/R_r$ times the squared coefficient norm. It follows that the signed evaluation form is bounded below by $(2T_r^2)^{-1} \|\text{coef}(P)\|^2$ on all complex polynomials of degree at most $2r$. Taylor truncation at that degree and the identical square completion give

$$Q_Z(f, f) \geq -D_r L(\beta L)^{4r+2} \|f\|_2^2 \quad (\beta L \leq \max(1, rR_r)^{-1})$$

for a finite constant D_r . Since $4r + 3 = p_e$, this is the required small-margin obstruction; (2.8) gives the logarithmic obstruction. Both hold for all complex tests, not only odd ones.

The qualification “centered” is essential. If an original divisor is globally conjugation-symmetric and is moved by $-i\gamma$ to put a target at ordinate zero, its original conjugation becomes $w \mapsto \overline{w} - 2i\gamma$, not $w \mapsto \overline{w}$. Thus ordinary global conjugation does not justify (3.21) at a nonzero target height.

3.8. Fixed continuous counting profiles. The proof uses only an integer low-node budget, a finite intermediate budget, and a polynomial tail. Consequently, for each fixed continuous nondecreasing $g : [1, \infty) \rightarrow [1, \infty)$, with $g(1) = 1$ and $g(2t) \leq \Delta g(t)$, both laws hold under $N_0(Y) \leq C(1 + Y)g(1 + Y)$, with constants depending on a, C, g . Indeed, continuity supplies $C(1 + \delta)g(1 + \delta) < \lfloor C \rfloor + 1$, and doubling gives $g(t) \leq \Delta t^{\log_2 \Delta}$. The finite lower examples still satisfy the entire count because their total mass is at most C . A common doubling constant does not supply a common modulus of continuity at one, so this observation asserts no uniformity over all such profiles.

4. EXACT FINITE SCREENING AND FIXED-BACKGROUND TRANSITIONS

We next determine an exact small-displacement coefficient for a symmetric finite screen and show that it survives every fixed polynomial-count axis background. The background may define an unbounded positive form on a fixed window. The argument does not assume its boundedness and does not let either the screen size or the background vary with the displacement. The finite-dimensional flat-limit and Schur-complement tools used below are established perturbation methods; see [7]. We give the signed calculation and the smooth-background transfer in full.

4.1. The coefficient and the two asymptotic conclusions. For an integer $n \geq 1$, define the divisor

$$(4.1) \quad \mathcal{S}_{n,\beta} = 128 \sum_{j=1}^n \left[i\beta n \cos \frac{(2j-1)\pi}{2n} \right] + [\beta] + [-\beta], \quad \beta > 0.$$

Here $[z]$ is an atom of unit multiplicity. Thus the screen has total multiplicity $128n + 2$, although only $n + 2$ distinct locations. Put

$$(4.2) \quad r = \lfloor n/2 \rfloor, \quad d = 2r + 1, \quad t_j = n \cos \frac{(2j-1)\pi}{2n} > 0 \quad (1 \leq j \leq r), \quad \alpha_n = \operatorname{arsinh}(1/n).$$

An empty sum is zero and an empty product is one. Define

$$(4.3) \quad \kappa_n = \frac{4}{n} \sum_{k=0}^{r-1} \sinh^2((2k+1)\alpha_n) = \frac{1}{n} \left\{ \frac{\sinh(4r\alpha_n)}{\sinh(2\alpha_n)} - 2r \right\},$$

$$(4.4) \quad P_n = \prod_{j=1}^r (1 + t_j^2) = n^n 2^{1-n} \begin{cases} \cosh(n\alpha_n), & n \text{ even}, \\ \sinh(n\alpha_n), & n \text{ odd}, \end{cases}$$

$$(4.5) \quad h_j = \frac{2^{2j+1}(j!)^4}{(2j+1)((2j)!)^2}.$$

In particular, $P_1 = 1$, $\kappa_1 = 0$, and $0 \leq \kappa_n < 16 < 128$. The positive coefficient governing the screen is

$$(4.6) \quad c_n = \frac{2P_n^2 h_d}{(d!)^2(1 - \kappa_n/128)} = \frac{2^{2d+2}(d!)^2 P_n^2}{(2d+1)((2d)!)^2(1 - \kappa_n/128)}.$$

On $L^2(-L, L)$, the finite-screen operator has exactly one negative eigenvalue, denoted by $\mu_-(n, \beta, L)$. For each fixed n , with $\epsilon = \beta L$,

$$(4.7) \quad \mu_-(n, \beta, L) = -Lc_n \epsilon^{2d} (1 + O_n(\epsilon^2)), \quad \epsilon \longrightarrow 0^+.$$

The estimate depends on β, L only through the displayed factor L and the parameter ϵ . Its error is not claimed to be uniform in n . The leading power in the magnitude of the negative eigenvalue is

$$2d = \begin{cases} 2n, & n \text{ odd}, \\ 2n + 2, & n \text{ even}. \end{cases}$$

After unitary scaling to $(-1, 1)$, a normalized negative eigenfunction tends, up to a unit scalar, to the normalized odd Legendre polynomial of degree d .

Now let $\mathcal{A}$ be a fixed locally finite divisor on $i\mathbb{R}$, with positive integer multiplicities and

$$(4.8) \quad N_{\mathcal{A}}(Y) := \sum_{\substack{i\gamma \in \mathcal{A} \\ |\gamma| \leq Y}} \nu_{\mathcal{A}}(i\gamma) \leq C_{\mathcal{A}}(1 + Y)^{m_{\mathcal{A}}}, \quad Y \geq 0,$$

for finite constants $C_{\mathcal{A}} \geq 0$ and $m_{\mathcal{A}} \geq 0$. The divisor may be empty, may contain the origin, and need not be separated at large ordinates. Set

$$Z_{\beta} = \mathcal{A} \sqcup \mathcal{S}_{n,\beta}, \quad q_n = \frac{2d}{2d+1},$$

where the union adds multiplicities, and define

$$(4.9) \quad \mathcal{M}_{Z_{\beta}}(L) = \sup_{0 \neq f \in C_c^{\infty}((-L, L))} \max \left\{ 0, -\frac{Q_{Z_{\beta}}(f, f)}{\|f\|_2^2} \right\}.$$

For fixed n and fixed $\mathcal{A}$,

$$(4.10) \quad \mathcal{M}_{Z_\beta}(x\beta^{-q_n}) \longrightarrow c_n x^{2d+1} \quad \text{locally uniformly for } x \in (0, \infty), \quad \beta \longrightarrow 0^+.$$

In particular, for each fixed $\eta > 0$, let

$$L_\eta(\beta) = \inf \left\{ L > 0 : \exists 0 \neq f \in C_c^\infty((-L, L)), Q_{Z_\beta}(f, f) \leq -\eta \|f\|_2^2 \right\},$$

with the infimum of the empty set equal to $+\infty$. Then

$$(4.11) \quad L_\eta(\beta) \sim \left(\frac{\eta}{c_n} \right)^{1/(2d+1)} \beta^{-2d/(2d+1)}.$$

The same asymptotic holds if the defining inequality is strict. Thus neither separation nor the size of the fixed axis background enters the leading constant. No uniformity over varying backgrounds is asserted.

4.2. Scaling, parity, and exact inertia. The unitary change of variables

$$g(x) = \sqrt{L} f(Lx), \quad -1 < x < 1,$$

preserves the norm and satisfies $H_f(z) = \sqrt{L} H_g(Lz)$. Consequently the screen form on $(-L, L)$ is L times the screen form on $(-1, 1)$ with parameter $\epsilon = \beta L$. We work on $(-1, 1)$ until scaling back at the end.

The nonzero imaginary nodes occur in the pairs $\pm i\epsilon t_j$, with an additional zero node if n is odd. Pairing the two imaginary nodes expresses their contribution as a positive cosine square plus a positive sine square. The real pair contributes

$$2 |\langle g, \cosh(\epsilon x) \rangle|^2 - 2 |\langle g, \sinh(\epsilon x) \rangle|^2.$$

All features here are real. The even and odd subspaces are therefore orthogonal invariant subspaces. The even block is a sum of positive squares, including the constant square if there is a zero node. On the odd subspace the form is exactly

$$(4.12) \quad q_\epsilon(g) = 256 \sum_{j=1}^r |\langle g, \sin(\epsilon t_j x) \rangle|^2 - 2 |\langle g, \sinh(\epsilon x) \rangle|^2.$$

For every $\epsilon > 0$, the $r + 1$ odd features in (4.12) are linearly independent. Indeed their entire continuations are combinations of exponentials with distinct exponents $\pm i\epsilon t_j, \pm \epsilon$, and distinct exponentials are independent: differentiating an identically zero combination at zero through one less than the number of exponents gives an invertible Vandermonde system. The corresponding even features are independent by the same argument, adding exponent zero when necessary.

For completeness, if Φ is the synthesis map for finitely many independent features and J is their diagonal matrix of nonzero signed weights, then its operator is $\Phi J \Phi^*$. Writing $\Phi = ER$, where E is an isometry onto its range and R is invertible, shows that its nonzero inertia is that of RJR^* , hence that of J . Applied to the odd block this gives r positive eigenvalues and one negative eigenvalue. The even block has $r + 1$ positive eigenvalues for even n , and $r + 2$ for odd n . Thus the full screen has rank $n + 2$, positive inertia $n + 1$, negative inertia one, and an infinite-dimensional nullspace.

4.3. The signed moment matrix and its Schur complement. Write $d_k = 2k + 1$ for $0 \leq k \leq r$. The feature series are

$$\sin(\epsilon t_j x) = \sum_{k \geq 0} (-1)^k t_j^{2k+1} \frac{\epsilon^{2k+1} x^{2k+1}}{(2k+1)!}, \quad \sinh(\epsilon x) = \sum_{k \geq 0} \frac{\epsilon^{2k+1} x^{2k+1}}{(2k+1)!}.$$

Define the real $(r+1)$ -square matrix F , with columns indexed by $0, \dots, r$, by

$$F_{j,k} = (-1)^k t_j^{2k+1} \quad (1 \leq j \leq r), \quad F_{r+1,k} = 1.$$

Let $J_s = \text{diag}(256, \dots, 256, -2)$ and $A = F^T J_s F$. Its entries are

$$(4.13) \quad A_{k\ell} = 256(-1)^{k+\ell} \sum_{j=1}^r t_j^{2(k+\ell+1)} - 2.$$

Factoring t_j from the first r rows of F leaves the Vandermonde matrix at $-t_1^2, \dots, -t_r^2, 1$. Thus F is invertible and A has inertia $(r, 1)$.

For $r \geq 1$, let

$$V_{j,k} = (-1)^k t_j^{2k+1} \quad (1 \leq j \leq r, \quad 0 \leq k < r), \quad B = V^T V, \quad \mathbf{1} = (1, \dots, 1)^T.$$

The matrix V is invertible by the same Vandermonde argument, so B is positive definite. The leading r -square block of A is

$$A_{<r} = 256B - 2\mathbf{1}\mathbf{1}^T,$$

and it is positive definite precisely when $\mathbf{1}^T B^{-1} \mathbf{1} < 128$. We next verify that

$$\mathbf{1}^T B^{-1} \mathbf{1} = \kappa_n.$$

For an odd polynomial of degree at most $2r-1$, write

$$R(t) = \sum_{k=0}^{r-1} a_k T_{2k+1}(t/n).$$

Discrete Chebyshev orthogonality on the original n nodes gives

$$\sum_{j=1}^r |R(t_j)|^2 = \frac{n}{4} \sum_{k=0}^{r-1} |a_k|^2.$$

To see the normalization, every nonconstant Chebyshev polynomial of degree less than n has discrete squared norm $n/2$ on all n nodes, with distinct degrees orthogonal, and pairing the positive and negative nodes halves the squared norm of an odd polynomial. A possible zero node contributes nothing. Explicitly, with $\theta_j = (2j-1)\pi/(2n)$, the finite geometric sum gives $\sum_{j=1}^n \cos(q\theta_j) = 0$ for $1 \leq q \leq 2n-1$. Applying $2\cos u \cos v = \cos(u-v) + \cos(u+v)$ proves the claimed orthogonality and normalization, since $T_k(\cos \theta) = \cos(k\theta)$. Also,

$$T_{2k+1}(i/n) = i(-1)^k \sinh((2k+1)\alpha_n).$$

The squared norm of evaluation $R \mapsto R(i)$, for the positive-node norm, is therefore the first expression in (4.3). In the alternative basis $R(t) = \sum_{k < r} c_k (-1)^k t^{2k+1}$, its squared node norm is $c^* B c$ and $R(i) = i \mathbf{1}^T c$. The squared norm of the same functional is $\mathbf{1}^T B^{-1} \mathbf{1}$, as required. Summing $2\sinh^2 v = \cosh(2v) - 1$ in a finite geometric progression gives the second expression in (4.3). Because $\alpha_n \leq 1/n$, $2r \leq n$, and $(2r-1)\alpha_n < 1$ when $r \geq 1$,

$$0 \leq \kappa_n < \frac{4r}{n} \sinh^2(1) \leq 2 \sinh^2(1) < 2e^2 < 16 < 128.$$

For $r = 0$, the same conclusion holds with $\kappa_1 = 0$. This proves positive definiteness of every leading moment block before the last one.

To compute the last Schur complement, put

$$a(s) = \prod_{j=1}^r (s + t_j^2) = \sum_{k=0}^r a_k s^k, \quad a_r = 1.$$

Then

$$\sum_{k=0}^r a_k (-1)^k t_j^{2k+1} = 0, \quad \sum_{k=0}^r a_k = P_n.$$

If $c = -a_{<r}$, the last column of the positive-feature matrix is Vc . In the quadratic form of A , replace $(x, z) \in \mathbb{R}^r \times \mathbb{R}$ by $y = x + cz$. Its value becomes

$$256y^T B y - 2(\mathbf{1}^T y + P_n z)^2.$$

The minimum over y , with z fixed, has coefficient

$$\begin{aligned} \sigma_n &= -2P_n^2 - 4P_n^2 \mathbf{1}^T (256B - 211\mathbf{1}^T)^{-1} \mathbf{1} \\ &= -\frac{2P_n^2}{1 - \kappa_n/128} < 0. \end{aligned}$$

For the second equality, solving the rank-one perturbation gives

$$(256B - 211\mathbf{1}^T)^{-1} \mathbf{1} = \frac{B^{-1} \mathbf{1}}{256(1 - \kappa_n/128)}.$$

Thus

$$(4.14) \quad \text{Schur}(A | A_{<r}) = \sigma_n = -\frac{2P_n^2}{1 - \kappa_n/128}.$$

When $r = 0$, this means simply $A = (-2)$.

The product formula (4.4) is also explicit. For even $n = 2r$, factor

$$T_n(z) = 2^{n-1} \prod_{j=1}^r (z^2 - (t_j/n)^2)$$

and evaluate at $z = i/n$, using $T_n(i/n) = (-1)^r \cosh(n\alpha_n)$. For odd $n = 2r + 1$, the corresponding factorization has the additional factor z , and $T_n(i/n) = i(-1)^r \sinh(n\alpha_n)$. These two evaluations give (4.4), including $n = 1$.

4.4. A nonsingular feature basis and the eigenvalue expansion. Let $\phi(\epsilon)$ be the column vector of the r sine features followed by $\sinh(\epsilon x)$, and put

$$D_\epsilon = \text{diag}_{0 \leq k \leq r} \frac{\epsilon^{d_k}}{d_k!}, \quad \psi(\epsilon) = D_\epsilon^{-1} F^{-1} \phi(\epsilon).$$

The apparent singularity at zero is removable. In fact the first $r + 1$ odd Taylor coefficients of $F^{-1} \phi$ vanish except at their selected component. The remaining series converge in $L^2(-1, 1)$, and show

$$(4.15) \quad \psi_k(\epsilon, x) = x^{d_k} + O_{n, L^2}(\epsilon^{2(r+1-k)}).$$

Each component extends analytically in the real parameter ϵ^2 through zero. If Ψ_ϵ denotes synthesis from these columns, the identity $\phi = F D_\epsilon \psi$ gives the exact odd-block operator

$$(4.16) \quad \Psi_\epsilon D_\epsilon A D_\epsilon \Psi_\epsilon^*.$$

Apply Gram–Schmidt to the columns of Ψ_ϵ in increasing degree:

$$\Psi_\epsilon = E_\epsilon R_\epsilon.$$

Here E_ϵ has orthonormal columns, and R_ϵ is upper triangular with positive diagonal. The limiting columns are independent odd monomials. The finitely many limiting squared Gram–Schmidt norms are therefore positive; the subtraction, division, and positive square-root operations show that E_ϵ and R_ϵ are analytic in ϵ^2 near zero.

Let p_j be the monic Legendre polynomial of degree j on $(-1, 1)$. The usual degree- j Legendre polynomial has leading coefficient $(2j)!/(2^j(j!)^2)$ and squared norm $2/(2j+1)$. Dividing by the square of this leading coefficient gives $\|p_j\|_2^2 = h_j$ in (4.5). Gram–Schmidt on the odd monomials consequently gives

$$(R_0)_{kk} = \sqrt{h_{d_k}}, \quad (E_0)_k = p_{d_k}/\sqrt{h_{d_k}}.$$

Set $V_\epsilon = D_\epsilon^{-1} R_\epsilon D_\epsilon$. For $j < k$, its (j, k) entry is that of R_ϵ times $\epsilon^{2(k-j)} d_j! / d_k!$; entries below the diagonal vanish. Therefore

$$V_\epsilon = \text{diag}(\sqrt{h_{d_0}}, \dots, \sqrt{h_{d_r}}) + O_n(\epsilon^2).$$

In the orthonormal coordinates E_ϵ , the operator (4.16) becomes

$$(4.17) \quad M_\epsilon = D_\epsilon C_\epsilon D_\epsilon, \quad C_\epsilon = V_\epsilon A V_\epsilon^T = \text{diag}(\sqrt{h_{d_k}}) A \text{diag}(\sqrt{h_{d_k}}) + O_n(\epsilon^2).$$

All errors here concern fixed finite matrices at fixed n .

For $r \geq 1$, write

$$M_\epsilon = \begin{pmatrix} B_\epsilon & u_\epsilon \\ u_\epsilon^T & a_\epsilon \end{pmatrix}, \quad S_\epsilon = a_\epsilon - u_\epsilon^T B_\epsilon^{-1} u_\epsilon.$$

The leading block of C_ϵ , and hence B_ϵ , is positive definite for small positive ϵ . Taking the Schur complement in (4.17) and using (4.14) gives

$$(4.18) \quad S_\epsilon = \frac{\epsilon^{2d}}{(d!)^2} \{h_d \sigma_n + O_n(\epsilon^2)\} < 0.$$

Indeed diagonal scaling factors out the final diagonal square, and the Schur complement of the limiting C_ϵ is $h_d \sigma_n$.

Also

$$(4.19) \quad B_\epsilon^{-1} u_\epsilon = O_n(\epsilon^2).$$

To verify this, write $D_<$ for the leading diagonal block of D_ϵ and $c_<$ for the upper part of the last column of C_ϵ . The vector in question is

$$D_<^{-1} (C_{\epsilon, <})^{-1} c_< \frac{\epsilon^d}{d!}.$$

The middle factors are uniformly bounded. Its coordinate with index $k < r$ is thus $O_n(\epsilon^{d-d_k})$, and $d - d_k \geq 2$.

Completing the square yields

$$(x, z)^T M_\epsilon(x, z) = (x + B_\epsilon^{-1} u_\epsilon z)^T B_\epsilon (x + B_\epsilon^{-1} u_\epsilon z) + S_\epsilon z^2.$$

Since $S_\epsilon < 0$, this is bounded below by $S_\epsilon(\|x\|^2 + |z|^2)$. Taking $x = -B_\epsilon^{-1} u_\epsilon z$ gives the other Rayleigh bound. Thus the least eigenvalue $\lambda_-(\epsilon)$ obeys

$$S_\epsilon \leq \lambda_-(\epsilon) \leq \frac{S_\epsilon}{1 + \|B_\epsilon^{-1} u_\epsilon\|^2} = S_\epsilon(1 + O_n(\epsilon^4)).$$

For $r = 0$, the matrix is scalar and (4.18) gives its eigenvalue directly. Combining these estimates with (4.14) proves

$$\lambda_-(\epsilon) = -c_n \epsilon^{2d} (1 + O_n(\epsilon^2)).$$

Unitary scaling proves (4.7).

The eigenfunction limit also follows from these bounds. The matrix is real symmetric and its negative eigenvalue is simple, so a real normalized negative eigenvector may be chosen. For $r \geq 1$, such an eigenvector (x, z) satisfies

$$x = -(B_\epsilon - \lambda_-(\epsilon)I)^{-1} u_\epsilon z.$$

Because B_ϵ is positive definite and $\lambda_-(\epsilon) < 0$, the operator $(I - \lambda_-(\epsilon)B_\epsilon^{-1})^{-1}$ has norm at most one. Equation (4.19) therefore gives $x = O_n(\epsilon^2)|z|$. After choosing the sign, $z \rightarrow 1$, and the limiting eigenfunction is the last column of E_0 , namely $p_d/\sqrt{h_d}$. The same conclusion is immediate when $r = 0$. All nonzero eigenvalues are captured by these finite coordinates. Density of $C_c^\infty((-L, L))$ in $L^2(-L, L)$, and boundedness of the finite-screen operator, show that its Rayleigh infimum on the compact smooth class is the same negative eigenvalue. In particular, with n fixed, the exact negative envelope is

$$(4.20) \quad e_{n,\beta}(L) := \mathcal{M}_{\mathcal{S}_{n,\beta}}(L) = -\mu_-(n, \beta, L) = c_n \beta^{2d} L^{2d+1} (1 + O_n((\beta L)^2)) \quad (\beta L \rightarrow 0).$$

For every positive β, L it is finite and strictly positive. It is nondecreasing in L , since the compact smooth test classes increase with the window. The supremum need not be attained by a compact smooth test, but such unit tests approach it arbitrarily closely. For even n , the degree is $d = n + 1$, so the powers in (4.20) are $2n + 2$ and $2n + 3$, respectively.

For reference, the first coefficients are

$$(4.21) \quad c_1 = \frac{4}{3}, \quad c_2 = \frac{1024}{44625}, \quad c_3 = \frac{23064}{151025}.$$

For $n = 1$ the odd block is rank one and its negative eigenvalue is exactly

$$2L - \frac{\sinh(2\beta L)}{\beta} = -\frac{4}{3}\beta^2 L^3 + O(\beta^4 L^5).$$

For $n = 2$, $t_1^2 = 2$, $P_2 = 3$, and $\kappa_2 = 1/2$. For $n = 3$, $t_1^2 = 27/4$, $P_3 = 31/4$, and $\kappa_3 = 4/27$. Substitution in (4.6) gives the remaining values in (4.21).

4.5. Smooth finite-dimensional recovery of the coefficient. To add a possibly unbounded positive axis form, we use tests from a fixed finite-dimensional smooth space rather than approximating an eigenfunction in the graph norm of that axis form. Let

$$\mathcal{V} = \text{span}\{x, x^3, \dots, x^d\} \subset L_{\text{odd}}^2(-1, 1).$$

Choose even cutoffs $\chi_\delta \in C_c^\infty((-1, 1))$, with $0 \leq \chi_\delta \leq 1$, tending pointwise to one as $\delta \rightarrow 0^+$, and put $\mathcal{H}_\delta = \chi_\delta \mathcal{V}$. For sufficiently small δ , these spaces have dimension $r + 1$. Indeed their basis vectors converge in L^2 to the independent monomials. If P_δ denotes orthogonal projection onto $\mathcal{H}_\delta$, best approximation gives

$$(4.22) \quad \left\| P_\delta x^{d_k} - x^{d_k} \right\|_2 \leq \left\| \chi_\delta x^{d_k} - x^{d_k} \right\|_2 \rightarrow 0, \quad 0 \leq k \leq r.$$

The projected moment vectors are consequently independent for small δ .

Fix one such space $\mathcal{H} = \mathcal{H}_\delta$, and let $P_\mathcal{H}$ be its projection. Compress (4.12) to $\mathcal{H}$. Its feature vectors are $P_\mathcal{H}\phi_j(\epsilon)$, and the exact factorization (4.16) holds with Ψ_ϵ replaced by $P_\mathcal{H}\Psi_\epsilon$. Their nonsingular limiting columns are $P_\mathcal{H}x^{d_k}$, by (4.15). Gram–Schmidt on these columns gives a positive diagonal $\sqrt{h_{k,\mathcal{H}}}$, where $h_{k,\mathcal{H}}$ is the squared distance of $P_\mathcal{H}x^{d_k}$ from the span of the earlier projected moments. The triangular scaling used in (4.17) therefore gives, explicitly,

$$C_{\epsilon,\mathcal{H}} = \text{diag}(\sqrt{h_{k,\mathcal{H}}}) A \text{diag}(\sqrt{h_{k,\mathcal{H}}}) + O_{n,\mathcal{H}}(\epsilon^2).$$

Its leading r -square block is positive definite, and its last Schur complement is $h_{r,\mathcal{H}}\sigma_n + O_{n,\mathcal{H}}(\epsilon^2)$. The same diagonal powers yield $B_{\epsilon,\mathcal{H}}^{-1}u_{\epsilon,\mathcal{H}} = O_{n,\mathcal{H}}(\epsilon^2)$; the completing-square Rayleigh bounds above apply without change. The compressed operator thus has exactly one negative eigenvalue for small $\epsilon > 0$, equal to

$$(4.23) \quad -c_n(\mathcal{H})\epsilon^{2d}(1 + O_{n,\mathcal{H}}(\epsilon^2)), \quad c_n(\mathcal{H}) = \frac{2P_n^2\rho_\mathcal{H}}{(d!)^2(1 - \kappa_n/128)},$$

where

$$(4.24) \quad \rho_{\mathcal{H}} = \text{dist}_{L^2}^2 \left(P_{\mathcal{H}} x^d, \text{span}\{P_{\mathcal{H}} x, \dots, P_{\mathcal{H}} x^{d-2}\} \right) > 0.$$

For $r = 0$, the span in this expression is $\{0\}$.

Equation (4.22) implies entrywise convergence of the finite moment Gram matrices to their positive definite unprojected Gram matrix. In the distance formula, the inverse of the leading Gram block is therefore continuous, with the scalar interpretation when $r = 0$. It follows that

$$(4.25) \quad \rho_{\mathcal{H}_\delta} \longrightarrow h_d, \quad c_n(\mathcal{H}_\delta) \longrightarrow c_n.$$

Moreover $\rho_{\mathcal{H}} \leq h_d$: project the unprojected minimizing residual p_d , use it as an admissible residual in (4.24), and use the contractivity of orthogonal projection. Thus $0 < c_n(\mathcal{H}) \leq c_n$.

Every vector in the fixed space $\mathcal{H}$ has support in the same compact subinterval of $(-1, 1)$. All norms on this finite-dimensional space are equivalent. Consequently, for every integer $N \geq 1$, integration by parts N times, without boundary terms, gives a finite constant $C_{\mathcal{H}, N}$ such that

$$(4.26) \quad |H_g(it)| \leq C_{\mathcal{H}, N} |t|^{-N} \|g\|_2 \quad (t \neq 0, g \in \mathcal{H}).$$

In particular, this bound is uniform over any choice of normalized negative eigenvectors of the compressed screen. Uniformity as $\delta \rightarrow 0$ is neither needed nor asserted.

4.6. The fixed axis background vanishes on the scaled tests. Local finiteness of $\mathcal{A}$ implies that its nonzero ordinate magnitudes are bounded away from zero. Choose an integer $N \geq 1$ with $2N > \max\{m_{\mathcal{A}}, 1\}$. Then

$$(4.27) \quad S_{\mathcal{A}, N} = \sum_{\substack{i\gamma \in \mathcal{A} \\ \gamma \neq 0}} \nu_{\mathcal{A}}(i\gamma) |\gamma|^{-2N} < \infty.$$

Indeed the finitely many nonzero ordinates of magnitude at most one give a finite sum. On $2^j < |\gamma| \leq 2^{j+1}$, (4.8) bounds the remaining contribution by a constant times $2^{j(m_{\mathcal{A}} - 2N)}$; the resulting geometric series converges. If there are no nonzero ordinates, the sum in (4.27) is zero.

For a unit vector $g \in \mathcal{H}$ define

$$f_L(u) = L^{-1/2} g(u/L).$$

This is a unit compact smooth test with support strictly inside $(-L, L)$, and

$$H_{f_L}(i\gamma) = \sqrt{L} H_g(iL\gamma).$$

Oddness gives $H_{f_L}(0) = 0$, so any multiplicity at the origin contributes nothing. Equations (4.26) and (4.27) give the uniform bound

$$(4.28) \quad 0 \leq Q_{\mathcal{A}}(f_L, f_L) \leq C_{\mathcal{H}, N}^2 S_{\mathcal{A}, N} L^{1-2N} \longrightarrow 0 \quad (L \longrightarrow \infty).$$

No boundedness of the axis form on the full L^2 space is being assumed.

We can now prove (4.10). Axis positivity and the finite-screen eigenvalue imply

$$(4.29) \quad 0 \leq \mathcal{M}_{Z_\beta}(L) \leq -\mu_-(n, \beta, L).$$

This in particular proves finiteness of $\mathcal{M}_{Z_\beta}(L)$, even if the positive axis form is unbounded on $L^2(-L, L)$. At $L = x\beta^{-q_n}$,

$$\epsilon = \beta L = x\beta^{1/(2d+1)}, \quad L\epsilon^{2d} = x^{2d+1}.$$

For any compact interval $I \subset (0, \infty)$, (4.7) and (4.29) therefore give

$$(4.30) \quad \mathcal{M}_{Z_\beta}(x\beta^{-q_n}) \leq c_n x^{2d+1} + O_{n, I}(\beta^{2/(2d+1)}) \quad (x \in I).$$

For the reverse bound, take $g = g_{\epsilon, \mathcal{H}}$ to be any normalized negative eigenvector of the compressed screen. Its scaled test has, by (4.23),

$$Q_{\mathcal{S}_{n,\beta}}(f_L, f_L) = -c_n(\mathcal{H})x^{2d+1} + O_{n,\mathcal{H},I}(\beta^{2/(2d+1)}).$$

On I , the right side of (4.28) is uniformly $O_{\mathcal{H},\mathcal{A},I}(\beta^{q_n(2N-1)})$. Hence

$$(4.31) \quad \mathcal{M}_{Z_\beta}(x\beta^{-q_n}) \geq c_n(\mathcal{H})x^{2d+1} - o_{n,\mathcal{H},\mathcal{A},I}(1) \quad (x \in I).$$

Given an error tolerance, first choose $\mathcal{H}_\delta$ so that $(c_n - c_n(\mathcal{H}_\delta)) \max_{x \in I} x^{2d+1}$ is smaller than that tolerance, using (4.25). With this smooth space fixed, let $\beta \rightarrow 0$ in (4.30) and (4.31). This proves the locally uniform limit (4.10). The order of the two limits avoids any assertion of uniform derivative bounds as the cutoffs approach the endpoints.

4.7. Inversion to the first prescribed-margin radius. The test classes increase with L , so $\mathcal{M}_{Z_\beta}(L)$ is nondecreasing. Put

$$x_\eta = (\eta/c_n)^{1/(2d+1)}.$$

For any $0 < x_- < x_\eta < x_+$, (4.10) implies, for sufficiently small β ,

$$\mathcal{M}_{Z_\beta}(x_- \beta^{-q_n}) < \eta, \quad \mathcal{M}_{Z_\beta}(x_+ \beta^{-q_n}) > \eta.$$

The first inequality, together with monotonicity, excludes the requested margin at every radius at most $x_- \beta^{-q_n}$. The second supplies an actual smooth test with margin strictly greater than η : a supremum strictly above η must have a member above η . In particular the defining set for $L_\eta(\beta)$ is nonempty for small β , and

$$x_- \leq \beta^{q_n} L_\eta(\beta) \leq x_+.$$

Let $x_- \uparrow x_\eta$ and $x_+ \downarrow x_\eta$. This proves (4.11) for both strict and non-strict definitions, without an attainment assumption.

For $n = 1$, the formula reads

$$L_\eta(\beta) \sim (3\eta/4)^{1/3} \beta^{-2/3}.$$

For $n = 2$ and $n = 3$, the exponent is $6/7$, with the different constants in (4.21). In general the radius exponent is $2n/(2n+1)$ for odd n , and $(2n+2)/(2n+3)$ for even n .

The fixed-background conclusion includes every fixed axis divisor satisfying (4.8). It does not include backgrounds whose locations or multiplicities vary with β . Likewise the fixed- n expansion (4.7) supplies no error bound uniform in a growing screen size. The result here is the exact prescribed-margin transition for a fixed configuration, not an assertion uniform in a growing count budget.

5. AMPLIFICATION AND GROWTH FOR ONE FIXED DIVISOR

We now keep the entire divisor fixed while the support window grows. Assume that Z is reflected, lies in $|\Re z| \leq a$, has positive integer multiplicities, and satisfies

$$(5.1) \quad N_0(Y) \leq C(1+Y)^m \quad (Y \geq 0), \quad C \geq 1, \quad m \geq 0.$$

Suppose that its off-axis part is nonempty, and put

$$(5.2) \quad s_Z = \sup\{\Re z : z \in Z, \Re z > 0\}, \quad \mathcal{M}_Z(L) = \sup_{0 \neq f \in C_c^\infty((-L,L))} \left(-\frac{Q_Z(f,f)}{\|f\|_2^2} \right)_+.$$

Thus $0 < s_Z \leq a$. The envelope is allowed to be infinite; we interpret $\log 0 = -\infty$ and $\log(+\infty) = +\infty$. The count hypothesis alone gives the lower rate

$$(5.3) \quad \liminf_{L \rightarrow \infty} \frac{\log \mathcal{M}_Z(L)}{L} \geq 2s_Z.$$

We first prove this assertion by retaining the free annihilation radius in the support–margin construction. We then give a weighted local condition under which the envelope is finite on every window and the lower rate is exact. No arithmetic information is used in this section.

5.1. Fixed-target amplification with arbitrarily small relative support loss. First suppose that a target has been centered at the real pair $\pm\beta$, with $0 < \beta \leq 1$. Denote the centered divisor by W , and fix an annihilation radius $R \geq 16(a+2)$. Apply the construction of (3.17) with $\ell = 1/\beta$ and $\tau = \beta T$, where T tends to infinity. In particular $\beta\ell = 1$, and $\tau \geq 1$ once $T \geq 1/\beta$. All counting budgets and all constants attached to this fixed R are fixed before T increases. The exponent estimate (3.10) has the form

$$(5.4) \quad K \leq a(T + \ell) + (M - 1)\{\log(1/\beta) + \log(1 + \beta T)\} + J_R,$$

where M is a fixed finite low-node budget and J_R is independent of T . The exact support estimate (3.15) consequently gives

$$(5.5) \quad L_T \leq \ell + T + 4K/R \leq (1 + 4a/R)T + A_R \log(2 + T) + B_R.$$

Here and below the finite constants may depend on the fixed target, the strip, the count, and R . This is not a uniform estimate over target heights or over growing values of R .

We spell out why the same construction gives a normalized, rather than merely unnormalized, exponential gain. Its local target values before translation are $\sqrt{\ell}$ and $-\sqrt{\ell}$. Multiplication by $\cosh(Tw)$ is the average of translations by T and $-T$, so it does not increase the L^2 norm. The chosen interpolation components have imaginary width $O((\ell + T)^{-1})$. Thus every selected reflected pair retains a negative contribution after translation, while all intermediate nodes vanish. The absolute tail is at most $\ell/2$. In particular the actual target, of multiplicity at least one, yields

$$(5.6) \quad Q_W(f_T, f_T) \leq -2\ell \cosh^2(\beta T) + \ell/2 \leq -\frac{\ell}{4}e^{2\beta T}.$$

These are the sign and tail conclusions of (3.17); they do not follow from translating an arbitrary previously negative test.

The physical-coordinate norm estimate in that construction is

$$(5.7) \quad \|f_T\|_2^2 \leq D_R(1 + \beta T)^{2M-2} \exp\left(\frac{D_R\beta^2 K}{(1 + \beta T)^2}\right).$$

The constant in the exponent can be enlarged to the same symbol D_R . Equation (5.4) shows that this exponent remains bounded as $T \rightarrow \infty$, and in fact tends to zero for a fixed target. Combining (5.6) and (5.7) therefore gives a nonzero compact smooth test for each sufficiently large T , with

$$(5.8) \quad -\frac{Q_W(f_T, f_T)}{\|f_T\|_2^2} \geq c_R \frac{e^{2\beta T}}{(1 + T)^{2M-2}}, \quad c_R > 0.$$

Its support is strictly inside the interval with radius bounded in (5.5), because the initial smoothing function has support strictly inside its prescribed interval.

Fix $0 < \delta < 1$ and, for each requested radius L , set

$$T = \frac{(1 - \delta)L}{1 + 4a/R}.$$

For every sufficiently large L , the logarithmic error in (5.5) is less than δL , and $T \geq 1/\beta$. The constructed test then lies in $C_c^\infty((-L, L))$. Equation (5.8) implies

$$\liminf_{L \rightarrow \infty} \frac{\log \mathcal{M}_W(L)}{L} \geq \frac{2\beta(1 - \delta)}{1 + 4a/R}.$$

First let $\delta \downarrow 0$, and then let the fixed radius R increase without bound. This proves a lower rate 2β . The order of limits matters: no control of A_R, B_R, c_R, D_R, J_R as $R \rightarrow \infty$ has been asserted or used. The construction supplies tests on every sufficiently large window, not just a subsequence.

For a target $\beta + i\gamma$ in Z , centering changes the count only to

$$(5.9) \quad N_\gamma(Y) \leq C(1 + |\gamma| + Y)^m \leq C(1 + |\gamma|)^m(1 + Y)^m.$$

If $W = Z - i\gamma$ and $f(u) = e^{-i\gamma u}g(u)$, then $H_f(z) = H_g(z - i\gamma)$, so $Q_Z(f, f) = Q_W(g, g)$, with the same norm and support. Hence the preceding conclusion applies at any fixed height.

The restriction $\beta \leq 1$ is immaterial for the growth statement. For $\alpha = \max\{a, 1\}$, let $\tilde{Z} = \{z/\alpha : z \in Z\}$, retaining multiplicities. Its strip width is at most one and its count is bounded by $C\alpha^m(1 + Y)^m$. If $f(u) = \sqrt{\alpha}g(\alpha u)$, direct substitution gives

$$(5.10) \quad \|f\|_2 = \|g\|_2, \quad H_f(z) = \alpha^{-1/2}H_g(z/\alpha), \quad Q_Z(f, f) = \alpha^{-1}Q_{\tilde{Z}}(g, g).$$

A radius αL for g becomes radius L for f ; the inverse scaling gives the exact envelope relation $\mathcal{M}_Z(L) = \alpha^{-1}\mathcal{M}_{\tilde{Z}}(\alpha L)$. Thus the lower rate for a target of displacement β/α scales back to 2β .

We have proved the lower rate for every fixed off-axis target. Taking their supremum proves (5.3), even when s_Z is not attained. In particular $\mathcal{M}_Z(L) > 0$ for every sufficiently large L , and it tends to infinity. For any finite requested margin, a supremum strictly greater than that margin contains an actual compact smooth test above it. Thus all finite margins are attained as inequalities on some finite window, without an eigenfunction-attainment assumption.

5.2. A weighted local condition sufficient for a finite envelope. Use one positive-real representative of each reflected pair, and assume

$$(5.11) \quad B_Z := \sup_{y \in \mathbb{R}} \sum_{\substack{z = \sigma + i\gamma \in Z \\ \sigma > 0, |\gamma - y| \leq 1}} \nu(z)\sigma^2 < \infty.$$

This condition permits infinitely many off-axis pairs, unbounded unweighted local counts when the displacements shrink, and arbitrarily close distinct locations. It holds for a finite off-axis part or a uniformly bounded unit-height count. The axis part requires no extra hypothesis beyond (5.1). We will prove, for every $L > 0$ and $0 < \varepsilon \leq 1$,

$$(5.12) \quad \mathcal{M}_Z(L) \leq 8B_Z(s_Z + \varepsilon)\varepsilon^{-2}L^2e^{2(s_Z + \varepsilon)L}.$$

In particular, for $L \geq 1$,

$$(5.13) \quad \mathcal{M}_Z(L) \leq 8e^2B_Z(s_Z + 1)L^4e^{2s_ZL}.$$

No optimality of the polynomial factor is asserted.

Fix a compact smooth test f supported in $(-L, L)$, and set

$$A_z = \frac{H_f(\sigma + i\gamma) + H_f(-\sigma + i\gamma)}{2}, \quad D_z = \frac{H_f(\sigma + i\gamma) - H_f(-\sigma + i\gamma)}{2}.$$

The contribution of the pair is exactly $2\nu(z)(|A_z|^2 - |D_z|^2)$; every axis term is nonnegative. The entire function

$$G_f(w) = \int_{\mathbb{R}} uf(u)e^{wu} du$$

satisfies

$$D_z = \frac{\sigma}{2} \int_0^1 \{G_f(t\sigma + i\gamma) + G_f(-t\sigma + i\gamma)\} dt.$$

Cauchy–Schwarz on this average gives

$$(5.14) \quad 2\nu(z)|D_z|^2 \leq \nu(z)\sigma^2 \int_0^1 \{|G_f(t\sigma + i\gamma)|^2 + |G_f(-t\sigma + i\gamma)|^2\} dt.$$

For fixed $t \in [0, 1]$, give each of $t\sigma + i\gamma$ and $-t\sigma + i\gamma$ the weight $\nu(z)\sigma^2$, and denote the resulting measure by μ_t . At $t = 0$, both weights are retained even though their locations coincide. Its mass in any radius-one vertical band is at most $2B_Z$. The area mean inequality for the subharmonic function $|G|^2$ yields

$$(5.15) \quad \begin{aligned} \int |G(w)|^2 d\mu_t(w) &\leq \frac{1}{\pi\varepsilon^2} \iint_{|\zeta-w|<\varepsilon} |G(\zeta)|^2 dA(\zeta) d\mu_t(w) \\ &\leq \frac{2B_Z}{\pi\varepsilon^2} \int_{-s_Z-\varepsilon}^{s_Z+\varepsilon} \int_{\mathbb{R}} |G(x+iy)|^2 dy dx. \end{aligned}$$

Indeed, for a fixed area point $x+iy$, all contributing centers have $|\gamma-y| < \varepsilon \leq 1$; their total weight is at most $2B_Z$. Their real parts lie in $[-s_Z, s_Z]$. Tonelli justifies the interchange of these nonnegative integrands, and no separation assumption enters the argument.

For $G = G_f$, Plancherel gives

$$\int_{\mathbb{R}} |G_f(x+iy)|^2 dy = 2\pi \int_{\mathbb{R}} u^2 |f(u)|^2 e^{2xu} du \leq 2\pi L^2 e^{2|x|L} \|f\|_2^2.$$

Discard the positive terms of Q_Z , sum (5.14), and apply (5.15). The real integral has length $2(s_Z + \varepsilon)$, and the t -integral has length one. This proves (5.12) with its displayed constant. Taking $\varepsilon = 1/L$ proves (5.13); taking $\varepsilon = 1$ proves finiteness also for smaller windows. The original form converges absolutely by (5.1). Only its nonnegative majorants were interchanged in the proof. There is no need to assert a bounded positive axis operator on the whole L^2 window.

5.3. Exact exponential rate and the first prescribed-margin radius. Combining (5.3) and (5.13) gives

$$(5.16) \quad \lim_{L \rightarrow \infty} \frac{\log \mathcal{M}_Z(L)}{L} = 2s_Z \quad \text{if } B_Z < \infty.$$

The logarithm is finite and ordinary for all sufficiently large L : finiteness follows from the weighted bound and positivity from the target construction. Define the first radius using actual tests by

$$(5.17) \quad r_Z(\eta) = \inf\{L > 0 : \exists 0 \neq f \in C_c^\infty((-L, L)), \quad Q_Z(f, f) \leq -\eta \|f\|_2^2\}, \quad \eta > 0.$$

It is finite, and

$$(5.18) \quad r_Z(\eta) \sim \frac{\log \eta}{2s_Z} \quad (\eta \rightarrow \infty).$$

Here the divisor, including all its multiplicities, is fixed.

To justify inversion without attainment, fix $0 < \delta < 2s_Z$. For all sufficiently large L , the rate law implies

$$e^{(2s_Z-\delta)L} \leq \mathcal{M}_Z(L) \leq e^{(2s_Z+\delta)L}.$$

At a radius strictly below $\log \eta / (2s_Z + \delta)$, the upper bound excludes a test of margin η . Bounded smaller radii are excluded for large η by monotonicity and finiteness of the envelope. At a radius strictly above $\log \eta / (2s_Z - \delta)$, the lower bound makes the supremum strictly exceed η , hence gives an actual test with strictly larger margin. Let the bounding radii approach their endpoints and then let $\delta \downarrow 0$. This proves (5.18), also when a strict margin inequality is used in (5.17).

The weighted condition is sufficient for the matching upper rate; it is not asserted to be necessary for that rate. It does not characterize all locally semibounded divisors. The lower rate (5.3) needs only the polynomial count, whereas a finite upper rate needs further control. In particular, local semiboundedness alone cannot be substituted for (5.11); the screened construction in Section 6 shows that fixed sparse counts can coexist with arbitrarily rapid growth of the finite negative envelope.

6. ARBITRARY GROWTH UNDER FIXED COUNTING BOUNDS

Finiteness of the negative envelope in each compact window does not by itself impose an exponential growth rate. This remains the case under both reflection and conjugation symmetry, with fixed linear origin counts and a fixed logarithmic unit-height bound. The finite screen calculation supplies the separation between two support radii needed to construct such an example.

More precisely, let $G : [2, \infty) \rightarrow (0, \infty)$ be any finite-valued nondecreasing function. There exists a locally finite divisor Z , with positive integer multiplicities, such that

$$(6.1) \quad Z = -\bar{Z} = \bar{Z} = -Z, \quad |\Re z| \leq 1 \quad (z \in Z), \quad [1] + [-1] \subset Z, \quad s_Z = 1,$$

$$(6.2) \quad N_0(Y) \leq Y + 2 \quad (Y \geq 0), \quad N_t(1) \leq 2 + \log(2 + |t|) \quad (t \in \mathbb{R}),$$

$$(6.3) \quad \mathcal{M}_Z(L) < \infty \quad (0 < L < \infty), \quad \mathcal{M}_Z(L) \geq G(L) \quad (L \geq 2).$$

The symmetries and counts include multiplicity. Here s_Z is the supremum of the positive real parts, and

$$\mathcal{M}_Z(L) = \sup_{0 \neq f \in C_c^\infty((-L, L); \mathbb{C})} \max \left\{ 0, -\frac{Q_Z(f, f)}{\|f\|_2^2} \right\}.$$

The series defining Q_Z is absolutely convergent on this test class. For each $L \geq 2$, a negative witness can be chosen real-valued and either even or odd; no fixed choice of parity is asserted for all windows. The construction also gives

$$(6.4) \quad \sup_{y \in \mathbb{R}} \sum_{\substack{z = \sigma + i\gamma \in Z \\ \sigma > 0, |\gamma - y| \leq 1}} \nu(z) \sigma^2 = \infty.$$

No continuity, differentiability, or computability assumption on G is used. The conclusion is domination of G , not an asymptotic formula for $\mathcal{M}_Z$.

6.1. Separating two radii with a finite screen. We use only the even values $n \geq 2$ of the finite screen (4.1), namely

$$\mathcal{S}_{n,\beta} = 128 \sum_{\ell=1}^n \left[i\beta n \cos \frac{(2\ell-1)\pi}{2n} \right] + [\beta] + [-\beta], \quad \beta > 0.$$

It has total multiplicity $128n + 2$ and is invariant under both conjugation and reflection. Write $e_{n,\beta}(L) = \mathcal{M}_{\mathcal{S}_{n,\beta}}(L)$. The exact finite-screen calculation in (4.20) gives, for fixed even n ,

$$(6.5) \quad e_{n,\beta}(L) = c_n \beta^{2d} L^{2d+1} (1 + O_n((\beta L)^2)), \quad d = n + 1, \quad c_n > 0.$$

Its error is not required to be uniform in n . The finite form is bounded on $L^2(-L, L)$, with exactly one negative eigenvalue. Consequently $e_{n,\beta}(L) > 0$, it is nondecreasing in L , and its value is approached by unit compact smooth tests. In particular, for each fixed $j \geq 1$ and even n ,

$$(6.6) \quad \frac{e_{n,\beta}(j)}{e_{n,\beta}(2j)} \longrightarrow 2^{-(2d+1)}, \quad e_{n,\beta}(j) \longrightarrow 0 \quad (\beta \downarrow 0).$$

Put $A_j = \max\{1, G(2j+2)\}$ and $\delta_j = 2^{-j}$. Choose an even integer $n_j \geq 2$, with $d_j = n_j + 1$, sufficiently large that

$$(6.7) \quad 2^{-(2d_j+1)} < \frac{\delta_j}{32A_j}.$$

This requires only that A_j be finite. Fixing n_j before decreasing β_j , choose $\beta_j > 0$ so that, with $e_j = e_{n_j, \beta_j}$,

$$(6.8) \quad \beta_j \leq \min\{2^{-j}, 1/n_j\}, \quad \frac{e_j(j)}{e_j(2j)} \leq \frac{\delta_j}{16A_j}, \quad e_j(j) \leq \frac{\delta_j}{4}.$$

Define the integer

$$(6.9) \quad m_j = \left\lceil \frac{4A_j}{e_j(2j)} \right\rceil.$$

The ceiling is controlled explicitly:

$$(6.10) \quad \begin{aligned} m_j e_j(j) &\leq 4A_j \frac{e_j(j)}{e_j(2j)} + e_j(j) \leq \frac{\delta_j}{2}, \\ m_j e_j(2j) &\geq 4A_j. \end{aligned}$$

The displacement can also be chosen to ensure

$$(6.11) \quad m_j \beta_j^2 \geq j.$$

Indeed, when n_j is fixed, (6.5) gives

$$\frac{4A_j \beta_j^2}{e_{n_j, \beta_j}(2j)} \sim \frac{4A_j}{c_{n_j}(2j)^{2d_j+1}} \beta_j^{2-2d_j} \longrightarrow \infty.$$

Since $d_j \geq 3$, this is compatible with all the smallness requirements in (6.8). The order here is important: impose all sufficiently-small- β_j requirements first, then define m_j by (6.9); m_j is not frozen while β_j is decreased.

Set

$$(6.12) \quad D_j = m_j \mathcal{S}_{n_j, \beta_j}, \quad W_j = m_j(128n_j + 2).$$

The real displacements and ordinate offsets of D_j are at most one in absolute value. Its total mass W_j is finite. By the smooth-test approximation for the finite screen, select

$$(6.13) \quad \phi_j \in C_c^\infty((-2j, 2j); \mathbb{C}), \quad \|\phi_j\|_2 = 1, \quad Q_{D_j}(\phi_j, \phi_j) < -2A_j.$$

For each stage the screen, its multiplicity, and this smooth test are now fixed. No uniform derivative estimate for the sequence of tests is needed.

6.2. A symmetric induction controlling both directions of interference. For a finite reflected divisor D , define

$$(6.14) \quad T_D(f) = \sum_{z \in D} \nu_D(z) |H_f(z)| |H_f(z^\#)|, \quad z^\# = -\bar{z}.$$

Thus $|Q_D(f, f)| \leq T_D(f)$; this quantity controls individual absolute terms and does not rely on cancellation within a component. For a fixed compact smooth function h , integration by parts yields, for every integer $N \geq 1$,

$$(6.15) \quad |H_h(\sigma + it)| \leq C_{h, N} (1 + |t|)^{-N} \quad (|\sigma| \leq 1, t \in \mathbb{R}).$$

For example, apply integration by parts to $e^{\sigma u} h(u)$; its derivatives have uniformly bounded integrals for $\sigma \in [-1, 1]$, and the bounded t -range is controlled directly. It follows that, for fixed finite D in this strip,

$$(6.16) \quad T_{D+i\Gamma}(h) \longrightarrow 0 \quad (|\Gamma| \rightarrow \infty), \quad T_D(e^{-i\Gamma u} h) \longrightarrow 0 \quad (|\Gamma| \rightarrow \infty).$$

These are finite sums with all multiplicities fixed before Γ tends to infinity.

Take $D_0 = [1] + [-1]$. We recursively choose positive centers γ_j and put

$$(6.17) \quad C_j^+ = D_j + i\gamma_j, \quad C_j^- = D_j - i\gamma_j, \quad E_j = C_j^+ \sqcup C_j^-, \quad f_j(u) = e^{-i\gamma_j u} \phi_j(u).$$

The notation $\sqcup$ adds divisor multiplicities. Because $D_j = \overline{D_j}$, conjugation exchanges the two components in E_j . At stage j , require

$$(6.18) \quad \begin{aligned} \gamma_1 &\geq 10, & \gamma_j &\geq \gamma_{j-1} + 10 \quad (j \geq 2), \\ \gamma_j &\geq 1 + 2 \sum_{i=1}^j W_i, & \log \gamma_j &\geq W_j, \end{aligned}$$

as well as

$$(6.19) \quad T_{E_j}(f_k) \leq 2^{-j-k-2} \quad (1 \leq k < j),$$

and

$$(6.20) \quad T_{D_0 \sqcup E_1 \sqcup \dots \sqcup E_{j-1}}(f_j) + T_{C_j^-}(f_j) \leq \frac{1}{4}.$$

All these requirements can be satisfied simultaneously. The counting conditions in (6.18) impose only a finite lower threshold on γ_j . Each term in (6.19) tends to zero by the first limit in (6.16), applied to the two translated copies of the fixed D_j and each fixed old test. The first term of (6.20) tends to zero by the second limit, with the old finite divisor fixed. The current opposite-height copy obeys the exact identity

$$(6.21) \quad T_{C_j^-}(f_j) = T_{D_j}(e^{-2i\gamma_j u} \phi_j) \longrightarrow 0 \quad (\gamma_j \rightarrow \infty).$$

Thus this last interaction has frequency separation $2\gamma_j$. There are only finitely many constraints at each stage, and f_j depends on the new center only through modulation of the already fixed ϕ_j . This proves the induction without a simultaneous infinite choice or a uniform bound on the smoothing constants.

Define

$$(6.22) \quad Z = D_0 \sqcup \bigsqcup_{j \geq 1} E_j.$$

Every component is finite, its centers tend to infinity, and the enclosing bands $[\pm\gamma_j - 1, \pm\gamma_j + 1]$ are disjoint. Therefore Z is locally finite. All multiplicities are positive integers. Imaginary translation preserves reflection since $(z + i\gamma)^\# = z^\# + i\gamma$; conjugation exchanges the two bands of each E_j . This proves all the symmetries in (6.1). The strip condition follows from the displacement choices, and the base pair realizes $s_Z = 1$.

6.3. Counts, including partially entered bands. Fix $Y \geq 0$. If there is an index with $\gamma_j - 1 \leq Y$, let k be the largest such index. Every component contributing to $N_0(Y)$ has index at most k . Each of its two bands contributes at most its full mass W_j , even if the interval has entered only part of the band. Consequently

$$(6.23) \quad N_0(Y) \leq 2 + 2 \sum_{j=1}^k W_j \leq 2 + \gamma_k - 1 \leq Y + 2.$$

If no such index exists, only the base pair can contribute, and the same bound holds. The use of the first edge $\gamma_k - 1$, rather than the center of a fully entered band, proves this at all endpoints as well.

A closed interval $[t-1, t+1]$ can meet at most one of the signed enclosing bands, since their centers are separated by at least ten. It cannot meet both such a band and the base ordinate zero. If it meets a band of index j , then $|t| \geq \gamma_j - 2$, so

$$(6.24) \quad N_t(1) \leq W_j \leq \log \gamma_j \leq \log(2 + |t|).$$

If it meets only the base ordinate, its count is at most two; otherwise its count is zero. This proves (6.2) with constants independent of G . The factor two in the origin-count budget accounts for both conjugate bands; it is not needed in the local count, whose interval meets at most one band.

6.4. Absolute convergence and local semiboundedness. The origin count implies

$$(6.25) \quad \sum_{z \in Z} \nu(z)(1 + |\Im z|)^{-2} < \infty.$$

Indeed the bounded central region is finite, and the part with $2^q < |\Im z| \leq 2^{q+1}$ is bounded by a constant times 2^{-q} . For any fixed compact smooth f , (6.15) bounds the absolute divisor summand by $C_f \nu(z)(1 + |\Im z|)^{-2}$. Hence the series for $Q_Z(f, f)$ converges absolutely, and may be grouped by (6.22). The same reasoning applies to the sesquilinear form on two compact smooth tests.

Fix a finite radius $L > 0$, and set $K = \max\{1, \lceil L \rceil\}$. For $j \geq K$, each of the two imaginary translations in E_j has negative envelope $m_j e_j(L)$. This follows from

$$H_{e^{i\gamma u} f}(z) = H_f(z + i\gamma),$$

because modulation preserves the norm and support. Since $L \leq j$, monotonicity and (6.10) imply

$$(6.26) \quad Q_{E_j}(f, f) \geq -2m_j e_j(L) \|f\|_2^2 \geq -\delta_j \|f\|_2^2 \quad (f \in C_c^\infty((-L, L))).$$

Thus the doubled negative budget of all sufficiently late components is summable.

The finitely many earlier components have finite negative envelopes. For the base pair, the even and odd features give

$$Q_{D_0}(f, f) = 2|\langle f, \cosh u \rangle|^2 - 2|\langle f, \sinh u \rangle|^2,$$

whose negative envelope is

$$e_0(L) = 2\|\sinh u\|_{L^2(-L, L)}^2 = \sinh(2L) - 2L.$$

Summing the lower bounds for finite unions and then using the absolute convergence just established yields

$$(6.27) \quad Q_Z(f, f) \geq -B_L \|f\|_2^2, \quad B_L = e_0(L) + 2 \sum_{j < K} m_j e_j(L) + \sum_{j \geq K} 2^{-j} < \infty.$$

This proves the first assertion in (6.3) for every finite L , including $L < 1$. No uniform convergence over the unit ball is required: the negative lower bounds are uniform and summable, while the original series converges absolutely separately on each smooth test. In particular, we do not assert that the full positive axis part is a bounded operator on $L^2(-L, L)$, or that a closed realization of the full infinite form has been constructed.

6.5. Survival of the prescribed witnesses. On its designated positive-height component the unit smooth test f_j satisfies exactly

$$(6.28) \quad Q_{C_j^+}(f_j, f_j) = Q_{D_j}(\phi_j, \phi_j) < -2A_j.$$

All old components, the base pair, and the current negative-height partner together cost at most $1/4$ in absolute value by (6.20). For each future stage $k > j$, its constraint (6.19) gives $T_{E_k}(f_j) \leq 2^{-k-j-2}$. Therefore

$$(6.29) \quad \sum_{k>j} T_{E_k}(f_j) \leq \sum_{k>j} 2^{-k-j-2} = 2^{-2j-2} \leq \frac{1}{16}.$$

Using absolute convergence and $A_j \geq 1$, we obtain

$$(6.30) \quad Q_Z(f_j, f_j) < -2A_j + \frac{1}{4} + \frac{1}{16} < -A_j.$$

The test is supported strictly inside $(-2j, 2j)$. For any $L \geq 2$, take $j = \lfloor L/2 \rfloor \geq 1$. Then $2j \leq L < 2j + 2$, and the increasing test classes and monotonicity of G give

$$\mathcal{M}_Z(L) \geq \mathcal{M}_Z(2j) \geq A_j \geq G(2j + 2) \geq G(L).$$

This proves domination at every radius $L \geq 2$, not merely along a sparse sequence.

We finish by checking the additional assertions. At height γ_j , the positive-real target $\beta_j + i\gamma_j$ has multiplicity m_j , so its contribution to the sum in (6.4) is $m_j \beta_j^2 \geq j$. This proves that assertion.

For real tests u, v , conjugation symmetry of Z makes $Q_Z(u, v)$ real. Together with Hermitian symmetry this gives

$$Q_Z(u + iv, u + iv) = Q_Z(u, u) + Q_Z(v, v), \quad \|u + iv\|_2^2 = \|u\|_2^2 + \|v\|_2^2.$$

Thus a complex negative test has a nonzero real or imaginary part with at least its negative Rayleigh margin. Invariance under $z \mapsto -z$ makes the form invariant under $Rf(u) = f(-u)$; its even and odd real subspaces are orthogonal for both the form and the norm. Splitting the chosen real test into those two parts therefore preserves the margin in at least one nonzero part. Normalization gives a real, definite-parity unit witness, without changing its strict compact support.

For instance, the choice $G(L) = e^{L^2}$ produces a divisor satisfying (6.1)–(6.2), with a finite bottom in every finite window, but

$$\lim_{L \rightarrow \infty} \frac{\log \mathcal{M}_Z(L)}{L} = +\infty.$$

Hence local semiboundedness and these fixed count bounds do not imply a finite exponential growth rate, even with full reflection and conjugation symmetry. The examples use multiplicities and are abstract divisors. No simple-divisor assertion, arithmetic realization, Euler product, or Riemann–von Mangoldt asymptotic is imposed.

7. THE ARITHMETIC INTERFACE

The preceding results separate information about a reflected divisor from information about an arithmetic formula. For clarity, we end by writing that interface in exactly the normalization used in this article.

Let $Z_\zeta = \{\rho - \frac{1}{2}\}$, where ρ runs over the nontrivial zeros of the Riemann zeta function, with multiplicity. The functional equation supplies reflection, and the classical ordinate count is polynomial. For $f \in C_c^\infty(-L, L)$, write

$$F_f(t) = H_f(it) = \int_{\mathbb{R}} f(u) e^{itu} du, \quad C_f(v) = \int_{\mathbb{R}} f(u) \overline{f(u-v)} du.$$

Thus $C_f(-v) = \overline{C_f(v)}$ and $C_f(0) = \|f\|_2^2$. The Guinand–Weil explicit formula, in additive coordinates, is

$$(7.1) \quad \begin{aligned} Q_{Z_\zeta}(f, f) = & \frac{1}{2\pi} \int_{\mathbb{R}} \left\{ \Re \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi \right\} |F_f(t)|^2 dt \\ & + 2\Re \left\{ H_f(1/2) \overline{H_f(-1/2)} \right\} \\ & - 2\Re \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} C_f(\log n). \end{aligned}$$

Here $\psi = \Gamma'/\Gamma$, and $\Lambda(n)$ equals $\log p$ if n is a positive power of a prime p and equals zero otherwise. See [3, Sections 2–4] for the explicit formula and its associated quadratic functional. Our formula follows by applying it to $f * \tilde{f}$, where $\tilde{f}(u) = \overline{f(-u)}$: its transform is $H_f(z) \overline{H_f(-\bar{z})}$, exactly the product in (2.3). The Fourier convention accounts for the factor $1/(2\pi)$.

Every term in (7.1) is well defined on the stated class. The prime sum is finite because $C_f(v) = 0$ for $|v| \geq 2L$. The Fourier transform decreases rapidly, while the real gamma multiplier grows like $\log |t|$ and is bounded below. The second line is a polar contribution, not a nonnegative form. In particular, for a real odd f it equals $-2|H_f(1/2)|^2$. Deleting that line or replacing it by a positive quantity would change the arithmetic problem.

Here is the exact implication supplied by the abstract lower-growth result. Suppose that, for one fixed $0 \leq \kappa < 1$ and every $\varepsilon > 0$, an independent arithmetic argument proves

$$(7.2) \quad \mathcal{M}_{Z_\zeta}(L) \leq C_\varepsilon e^{(\kappa+\varepsilon)L} \quad (L \geq 1).$$

If there is an off-axis zero, the lower rate in Section 5 gives

$$2 \sup_{\rho} \left| \Re \rho - \frac{1}{2} \right| \leq \liminf_{L \rightarrow \infty} \frac{\log \mathcal{M}_{Z_\zeta}(L)}{L} \leq \kappa.$$

If there is no off-axis zero, the same zero-location conclusion is immediate. Hence

$$(7.3) \quad \left| \Re \rho - \frac{1}{2} \right| \leq \kappa/2 \quad \text{for every nontrivial zero } \rho.$$

In particular, subexponential negative growth would imply the Riemann Hypothesis. The weighted sufficient condition of Section 5 is not needed for this implication; only the count-based lower rate is used.

The hypothesis (7.2) is not proved in this article. The signed prime sum, the gamma term, and the polar contribution would have to be controlled together, uniformly over every admissible test in the window. Counting bounds for a divisor and positivity of one finite compression do not provide that estimate. Conversely, the arbitrary-growth divisors of Section 6 are not asserted to be zero sets of arithmetic L -functions and do not refute an arithmetic inequality.

Several questions remain on the abstract side as well. The constants in the support–margin law are not uniform in a growing count budget; the exact leading minimax constant is undetermined. The weighted local bound is a sufficient condition for the matching upper rate; it is not asserted to be necessary. A characterization of growth that retains the compensation supplied by the axis part would be stronger than either a pure counting bound or an absolute estimate of the off-axis contribution. These are distinct questions from proving (7.2).

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NOTE ON AI USE

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing some ideas and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

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DEPARTMENT OF AEROSPACE ENGINEERING, INDIAN INSTITUTE OF SCIENCE
 Email address: arun.chandru@gmail.com